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REVIEW

Artificial Intelligence for Materials Discovery, Development, and Optimization

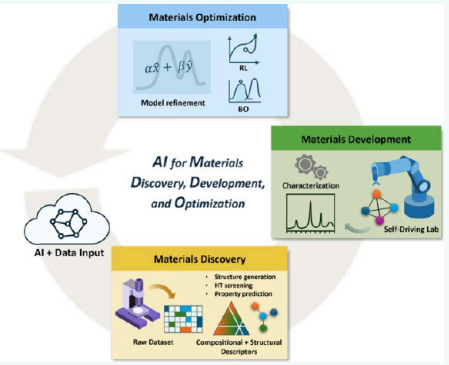
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ABSTRACT: This review highlights the recent transformative impact of artificial intelligence (AI), machine learning (ML), and deep learning (DL) on materials science, emphasizing their applications in materials discovery, development, and optimization. AI-driven methods have revolutionized materials discovery through structure generation, property prediction, high-throughput (HT) screening, and computational design while advancing development with improved characterization and autonomous experimentation. Optimization has also benefited from AI's ability to enhance materials design and processes. The review will introduce fundamental AI and ML concepts, including supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning (RL), alongside advanced DL models such as recurrent neural networks (RNNs), convolutional neural networks (CNNs), graph neural networks (GNNs), generative models, and Transformer-based models, which are critical for analyzing complex material data sets. It also covers core topics in materials informatics, including structure−property relationships, material descriptors, quantitative structure−property relationships (QSPR), and strategies for managing missing data and small data sets. Despite these advancements, challenges such as inconsistent data quality, limited model interpretability, and a lack of standardized data-sharing frameworks persist. Future efforts will focus on improving robustness, integrating causal reasoning and physics-informed AI, and leveraging multimodal models to enhance scalability and transparency, unlocking new opportunities for more advanced materials discovery, development, and optimization. Furthermore, the integration of quantum computing with AI will enable faster and more accurate results, and ethical frameworks will ensure responsible human-AI collaboration, addressing concerns of bias, transparency, and accountability in decision-making.



KEYWORDS: *autonomous experimentation, computational materials design, deep learning, generative models, high-throughput screening, machine learning, material property prediction, materials informatics, physics-informed AI*

INTRODUCTION

Historically, materials have been discovered by sheer coincidence or empirical experimentation. An illustrative instance is the discovery of vulcanized rubber, developed in the 19th century through the observation that heating with additives like sulfur can enhance the resilience of rubber.[1] Materials discoveries, which started with hands-on experiments, leaped with the development of theoretical frameworks like Kepler's Laws and Maxwell's equations.[2,3] However, simulations became essential for tackling scientific challenges as theoretical complexities grew. Simulations and a surge in experimental data have become prominent in modern science.[4−6] Today, scientists primarily rely on sophisticated instruments rather than traditional observation methods, such as manually operated optical telescopes. These instruments transmit data to centralized databases that researchers can access through computers. In this transformed scientific landscape, data are first captured by instruments or simulations and then processed by software before being stored as knowledge in computers. Scientists engage with their data at a later stage while processing. The methodologies and technologies of data-intensive science are distinct enough to be recognized as a separate paradigm from computational science, marking a new era in scientific exploration.[7,8]

The foundation of materials science and engineering lies in processing, structure, property, and performance.[9,10] The key

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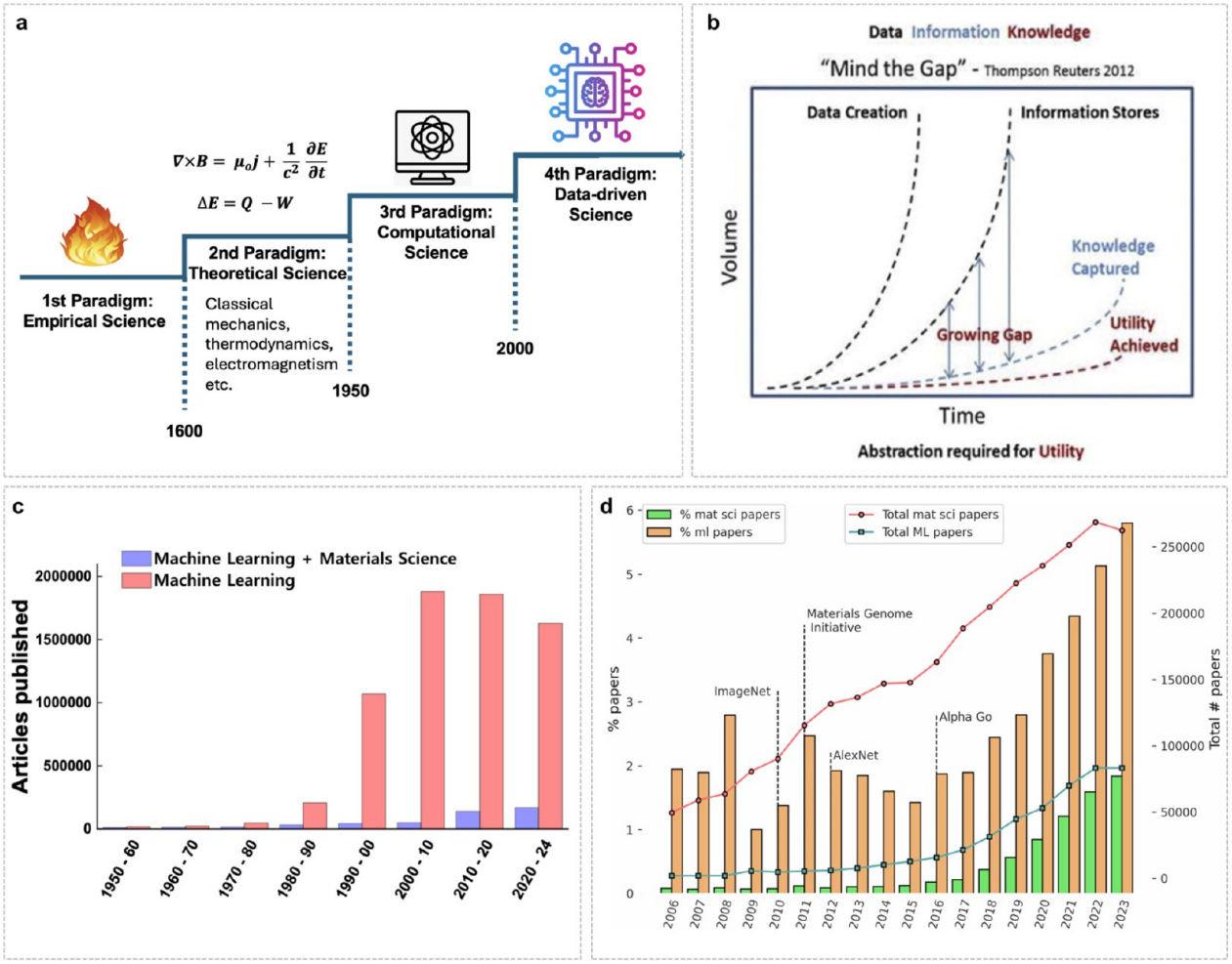


Figure 1. Illustrations of the shift toward data-driven science and the growing impact of ML in materials research. (a) The evolution of scientific paradigms from empirical and theoretical science to computational and data-driven approaches. (b) The increasing gap between data creation and knowledge extraction. Reprinted with permission from ref 30. Copyright 2013 Elsevier Inc. (c) The rise in ML and ML-materials science publications from 1950 to 2024 based on Google Scholar search results. (d) The growing percentage of materials science papers utilizing ML from 2008 to 2023, with key milestones like the MGI driving this trend. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 31. Copyright 2024 The Author(s).

to innovation is to find the interrelation among these four cornerstones.[11] The atomic arrangement defines the fundamental structure of a material; however, structure is inherently multiscale, with hierarchical features that span electronic, atomic, nano-, micro-, and macrolevels—all of which interact to govern the material's properties and, ultimately, its performance.[12−14] The structure of materials, in turn, governs key physical properties—including density, hardness, electrical resistivity, and thermal expansion—which can be systematically tuned by adjusting processing parameters.[9]

Engineered materials are highly customized and optimized for applications, offering enhanced performance, durability, or cost-effectiveness compared to naturally occurring or conventional materials.[15−17] Examples of engineered materials include high-strength alloys used in aerospace engineering, biodegradable polymers for medical implants, and advanced ceramics for thermal insulation in spacecraft.[9,11,18] As a result, continuous progress of materials research and development is of utmost importance.

Discovering, developing and optimizing engineered materials is a road with many challenges. Although a 20-year laboratory-to-market cycle has long been regarded as typical,[19] the U.S. Materials Genome Initiative (MGI) has demonstrated that integrated computational−experimental workflows can compress this timeline dramatically.[20] For example, the corrosion-resistant ultrahigh-strength steel *Ferrium S53* moved from concept to flight-qualified landing-gear components in fewer than six years, and *Ferrium M54* hook-shank steel was designed, certified, and first commercialized within four years. These cases illustrate how MGI principles can shorten materials deployment below the historic 20-year benchmark.[21,22] Additionally, once a material becomes integral to a technology, it tends to remain in use due to the significant investment required to establish widespread production capabilities, as evidenced by the enduring dominance of silicon in the semiconductor industry.[23] Hence, the selection of materials for targeted applications is crucial for ensuring the success of their integration, particularly as numerous emerging technological fields are now seeking materials with promising potential. Equally critical is a candidate's synthetic accessibility and manufacturability, that is, the ability to produce the material reproducibly, at scale, and at an acceptable cost and environmental footprint. In practice, this 'lab-to-fab' constraint often gates development, with many computationally attractive

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compounds proving infeasible to synthesize or integrate into existing process flows.[24−27]

In the early 21st century, exploring novel materials primarily relied on direct materials design approaches involving a combination of experimental assessments and theoretical simulations.[28] These methods, though effective, were time- and labor-intensive and heavily dependent on the expertise and intuition of researchers. The trial-and-error approach required prior knowledge to narrow the search space for materials with the desired properties. This constrained exploration to only familiar chemical spaces led to an incomplete mapping of the relationship between structure, composition, and properties.[29]

The shift, as outlined in Figure 1a, where scientific inquiry has transitioned from empirical observations to theoretical models, computational methods, and, most recently, data-driven approaches reflects the broader evolution of science. However, despite the drastic increase in data generation, as shown in Figure 1b, there remains a growing gap between the creation and storage of data and their transformation into meaningful knowledge and practical technologies.[30] This highlights the need for more efficient methods of extracting meaningful data from vast data sets.

The increasing role of machine learning (ML) in addressing these challenges is evident from the sharp rise in the number of ML-related publications, particularly in materials science, as depicted in Figure 1c. The integration of ML into materials research has gained momentum in recent decades, particularly after key milestones like the MGI,[31] as illustrated in Figure 1d. These developments underscore the necessity of adopting data-driven approaches to reduce discovery timelines and align with the accelerated pace of scientific and technological advancements.

Materials informatics has seen rapid growth recently, driven by the increasing amount of experimental and computational data.[32−36] Materials informatics uses mathematical and information science methods to leverage materials data to predict properties for new materials.[37,38] A critical aspect of this endeavor is developing a descriptor model capable of predicting the property of interest based on known material-specific features.[39] The quantitative structure−property relationship (QSPR) model in materials informatics is a significant descriptor model, utilizing material structural and compositional features as input variables.[40]

Traditional linear and nonlinear correlation methods often struggle to manage the complex relationships between inputs and outputs of material properties.[41−45] However, artificial intelligence (AI) advancements, such as ML, have enabled sophisticated modeling of more complex relationships in materials science.[46−51] Within this spectrum, deep learning (DL), a specialized form of ML, employs neural networks to unravel high-dimensional and complex data, offering profound capabilities for discovering intricate material properties, and thus guiding the materials discovery,[52] development,[53,54] and optimization.[55−58] Some successful applications of the techniques include predicting fatigue strength,[59] determining physical[60] and mechanical[61] properties, forecasting the electronic bandgap (E_g) of perovskite materials,[62] estimating catalytic activities[63] and acidity constants,[64] discovering high-temperature polymer dielectric materials for energy storage applications,[65] designing heterogeneous catalysts,[66] modeling precise structures of organic light-emitting diode materials,[67] predicting new superconductors,[68] and innovating photovoltaic material technologies for enhanced performances.[69]

These methodologies and technologies have driven the field of materials informatics.[70−72] Leveraging today's large materials data sets enables a shift to inverse design,[73−75] a data-driven loop that begins with a target property specification (*e.g.*, specific initial discharge capacity ≈ 200 mAh/g[76]) and, using ML models, iteratively proposes and refines candidate compositions or microstructures most likely to meet that target. Consequently, scientists can discover, develop, and optimize new materials directly from intelligent models trained on available data. With time, the application of AI and ML in materials science has become increasingly prominent (Figure 1c,d). While ML potentials (MLPs) represent a significant and widely used area, this review does not focus on that topic. Readers interested in MLPs are referred to recent dedicated reviews.[77−79]

This review explores how AI, including ML and DL, reshapes the discovery, development, and optimization of materials. It begins with foundational concepts and data strategies for addressing issues like missing or limited data sets. The review then examines AI-driven advances in materials discovery, such as property prediction, structure generation, and high-throughput (HT) screening; in materials development, where AI enables enhanced characterization, closed-loop synthesis, and autonomous experimentation; and in materials optimization, where AI accelerates design and process refinement through techniques such as reinforcement learning (RL) and Bayesian optimization (BO). Key challenges, including data inconsistency, model interpretability, and integration of multimodal data sets, are critically assessed. Finally, the review outlines future directions, such as causal reasoning, multimodal generative models, physics- and quantum-informed AI, and ethical AI-human collaboration, highlighting AI's growing potential to unify theory and experiment in materials research.

AI IN MATERIALS SCIENCE

AI transforms materials science by providing advanced tools for discovering, predicting, and optimizing material properties.[80] ML and DL techniques enable complex data analysis, while materials informatics integrates these approaches with large databases and predictive models to accelerate materials discovery.[81] Additionally, advanced AI methods such as transfer learning (TL),[82] data augmentation,[83] and emerging chain-of-thought (CoT) reasoning frameworks,[84] where a large language model (LLM) explicitly generates step-by-step rationales, further boost advancements on multistage scientific tasks while making model behavior more interpretable. Prompting strategies like CoT prompting,[85] Zero-Shot-CoT,[86] and Tree-of-Thought[87] have already demonstrated significant performance gains on logic-intensive benchmarks and are beginning to be applied to molecular structure elucidation, chemistry knowledge understanding, and synthesis plan and execution.[88−92] While CoT reasoning and prompting strategies specific to materials science are not covered in this review, interested readers are encouraged to consult other comprehensive reviews for further details.[88,93−95]

Basic Concept of AI. AI is dedicated to building systems that can carry out tasks normally requiring human intellect, such as identifying patterns, making decisions, and solving problems.[96] While AI is popularly viewed as a branch of computer science, several authors argue that modern AI methods are best interpreted as large-scale applications of statistical inference and optimization—that is, as a continuation of applied statistics itself.[97,98] At its core, AI builds mathematical models mimicking human cognition using principles like logic, probability, and optimization. Symbolic AI uses if-then rules (if X, then Y) primarily when domain knowledge can be encoded into rules.[96]

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AI systems can be grouped into distinct categories according to their functions and capabilities.[99−101] Diagnostic/descriptive AI analyzes historical data to identify patterns and root causes. Predictive AI forecasts future outcomes by using current and past data. Prescriptive AI provides the optimal course of action by providing recommendations based on data analysis. Generative/cognitive AI (GenAI) mimics human creativity and intelligence to automate and assist in content creation. Reactive AI responds to inputs with predefined actions without the ability to learn from past experiences. Limited memory AI learns from past experiences to inform current decisions and improve its performance over time. Advanced types include the theory of mind AI, aiming to understand human emotions, beliefs, and intentions, and self-aware AI, a theoretical concept with its own consciousness and self-awareness. Finally, narrow (weak) AI specializes in specific tasks but lacks generalization capabilities. At the same time, general (strong) AI mimics human intelligence across multiple domains, can perform knowledge transfer among different domains, and can adapt to new conditions autonomously. Most current AI applications fall under narrow (weak) AI.[99]

Fundamental ML Concepts. ML, a branch of AI, encompasses algorithms that enable computers to discover patterns or derive rules from data on their own without being explicitly programmed.[102] Historically, Arthur Samuel's 1959 checkers program crystallized the field—one of the first systems explicitly labeled 'machine learning.'[103] Consequently, ML both predates and underpins contemporary AI: many of today's AI breakthroughs are direct descendants of earlier ML research. ML is built on the idea that a model can generalize from examples and, in many (but not all) settings, improve its performance over time as it is exposed to more data; equally important, some widely used ML systems, such as the probabilistic base-caller for automated deoxyribonucleic acid (DNA) sequencing[104] or support vector machine diagnostic classifiers for breast cancer,[105] achieve high utility from a single static training set without ongoing updates. There are several ML types, including supervised learning,[106] unsupervised learning,[107] RL,[108] semi-supervised learning,[109] and evolutionary and optimization algorithms.[110−112]

In supervised learning, a model is taught using input−output examples with known labels so that it can infer the relationship mapping inputs to their corresponding outputs. In regression, the relationship between X and Y is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon \tag{1}$$

where β_0 is the intercept (or bias), β_1, β_2, ⋯, β_n are the coefficients, and ε is the error. The objective of this model is to reduce the sum of the squared differences between its predictions and the true values.[96] Classification, another widely used supervised learning technique, leverages logistic regression to estimate the probability P of an input belonging to a specific predefined class through a sigmoid function

$$P(Y = k|X) = \frac{1}{1 + e^{-(\beta_{0,k} + \beta_{1,k}X_1 + \cdots \beta_{n,k}X_n)}} \tag{2}$$

where $P(Y = k \mid X)$ represents the probability of the input X belonging to class k.[113] In materials science, supervised learning has been used to automate and enhance the efficiency of powder X-ray diffraction (XRD) analysis.[114]

Unsupervised learning focuses on identifying patterns in data without using input-output pairs. A common technique is clustering, in which data samples are clustered according to their shared characteristics without predefined labels. Another method, principal component analysis (PCA), reduces the data set's dimensionality by identifying the axes that explain the greatest variation. Both approaches reveal data hidden structures without supervision.[113] Unsupervised learning has been applied to discovering new materials, like superalloys, by analyzing sparse, unlabeled data in large, unexplored spaces.[115]

RL focuses on learning optimal actions through interaction with the environment.[116] The agent learns a policy $\pi(a|s)$, which specifies the probability of choosing an action a in a specific state s, with the objective of maximizing the cumulative reward over time. The cumulative reward is expressed as

$$R = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right] \tag{3}$$

where r_t is the reward at time step t, and γ is the discount factor that determines the importance of future rewards. To evaluate how good it is to be in a given state, RL employs the Bellman optimality equation

$$(V) = \max_a \mathbb{E}[r_t + \gamma V(s_{t+1})|s_t = s,\ a_t = a] \tag{4}$$

which recursively defines the state-value function as the maximum expected return achievable from the state.[116] Through successive updates of eq 4, the algorithm learns—via trial and error—the best actions over time. RL has been used to optimize the energy harvesting capability of a metamaterial system under realistic, stochastic excitations at the microscale.[117]

Semi-supervised learning integrates supervised and unsupervised approaches, leveraging insights from one to enhance the other.[109] For instance, unlabeled data can improve classification accuracy, while clustering can benefit from knowing whether specific data points belong to the same group. Semi-supervised learning has been employed in the materials science domain to categorize inorganic materials synthesis protocols described in plain-language text. This approach combines unsupervised clustering of experimental steps with guided annotation to link these steps to specific synthesis categories.[118]

Evolutionary algorithms and optimization algorithms such as BO are advanced ML techniques that can effectively solve optimization problems. Inspired by natural selection, evolutionary algorithms iteratively improve candidate solutions using operators like selection, crossover, and mutation.[112] Evolutionary algorithms have been utilized to identify and optimize materials by mathematically representing them as genomes. They are also used to describe fitness landscapes and employ mutation operators that are routinely used in materials science applications.[119] Meanwhile, BO, suitable for both maximization and minimization problems, models the objective function as a Gaussian process.[120] It selects the next sampling point by optimizing a selection criterion, for example, the expected improvement (EI) acquisition function. For maximization, EI is given by

$$\mathrm{EI}(x) = E[\max(0, f(x) - f(x^+))] \tag{5}$$

where $f(x^+)$ is the current best-observed value, and for minimization, the EI becomes

$$\mathrm{EI}(x) = E[\max(0, f(x^*) - f(x))] \tag{6}$$

where $f(x^*)$ is the minimum observed value. BO has been applied within materials science for a range of applications, including optimizing the light absorption properties of quasi-random solar cells and conducting combinatorial searches for optimal hybrid organic−inorganic perovskite materials designs.[121,122]

Fundamental DL Concepts. Unlike ML, which requires manual feature extraction, DL uses neural networks to automatically capture complex patterns and abstractions from data.[123] This makes DL highly effective for learning intricate data in the materials science domain.[124] Standard DL models include recurrent neural networks (RNNs), convolutional neural networks (CNNs), graph neural networks (GNNs), generative models, and Transformer-based models.

RNNs are designed for sequential data such as time series or text, maintaining a hidden state that updates at each time step to retain information from previous inputs. The hidden state h_t at time t is updated as

$$h_t = \mathcal{O}(W_h h_{t-1} + W_x x_t + b) \tag{7}$$

where W_h and W_x are weight matrices of hidden states and input data, respectively, x_t is the input data at the time t, and $\mathcal{O}$ is an activation

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function.[125] However, RNNs struggle with the vanishing gradient problem, making it difficult to learn long-term dependencies. Long short-term memory (LSTM) solves this issue with gating mechanisms that help retain information over extended periods.[125] RNN has been used in materials science to capture the time-dependent behavior of rate-dependent materials in a microscale analysis of heterogeneous materials.[126]

CNNs excel at processing spatial data, like images, by applying convolutional filters to detect local patterns and feature hierarchies.[127] This process is mathematically represented as

$$y_{i,j,k} = \sum_m \sum_n x_{i+m,j+n} w_{m,n,k} + b_k \tag{8}$$

where $y_{i,j,k}$ is the output at the position (i, j) for the kth filter, $x_{i+m,j+n}$ is the input value at the position $(i + m, j + n)$, $w_{m,n,k}$ represents the filter weights and b_k is the bias term. CNNs have proven highly effective in tasks such as image recognition due to their ability to learn hierarchical features from data automatically. For example, CNNs have been used to classify the composition and cycling states of Lithium Nickel Cobalt Manganese (NCM) Oxide cathode materials from SEM images.[128]

GNNs are designed for graph-structured data, making them effective in areas like crystal graph representation.[129] GNNs iteratively update node representations by aggregating information from neighboring nodes and their connections, capturing complex relationships within a graph.[130] The update of a node's feature vector h_v at iteration t is

$$h_v^{(t)} = \sigma(W \cdot \mathrm{AGGREGATE}(h_u^{(t-1)} | u \in N(v))) \tag{9}$$

where $h_u^{(t-1)}$ is the feature vector of the neighboring node u in the previous iteration, W is a learnable weight matrix, σ is a nonlinear activation function, and $N(v)$ represents the set of neighbors of a node v. This aggregation allows GNNs to learn node embeddings that reflect the graph's structural properties.

In addition to their predictive capabilities, DL models can generate new data samples from learned distributions. Therefore, advanced generative models have emerged, using different methods to create data, particularly in image and natural language generation.

One of the earliest and most notable generative models is the variational autoencoder (VAE). VAEs encode data into a latent space and then decode it back, regularizing the latent space to follow a distribution.[131] The critical optimization process for VAEs is based on maximizing the evidence lower bound (ELBO), described as

$$\log p(x) \geq \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{\mathrm{KL}}(q(z|x) \| p(z)) \tag{10}$$

where $p(x)$ is the probability of the data, $q(z \mid x)$ is the approximate posterior, $p(z)$ is the prior, and D_{KL} is the Kullback–Leibler divergence. By maximizing the ELBO, VAEs generate new, realistic data points. This model has been used to discover the physical and chemical phenomena from the electron, optical, and scanning microscopy data.[132]

Generative adversarial networks (GANs) are another class of generative models. In GANs, two networks—the generator (G) and the discriminator (D)—are trained competitively. The generator creates fake and generated data, while the discriminator differentiates between real and generated data. The following min-max game generation describes the optimization process.[133]

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\mathrm{data}}(x)}[\log D(x)] + \mathbb{E}_{z \sim p_z(z)}[\log(1 - D(G(z)))] \tag{11}$$

where $p_{data}(x)$ is the real data distribution and $p_z(z)$ is the distribution of the noise used to generate samples. GANs have been widely adopted to study composition design, processing optimization, crystal structure search, microstructure characterization, and defect detection.[134]

Diffusion models are an emerging class of generative models that have shown significant promise, especially in image generation tasks.[135] These models work by progressively adding noise to data and then learning to reverse this process to generate samples from pure noise. The gradual denoising process allows diffusion models to achieve impressive results in generating high-quality images, making them competitive with GANs in fields such as the generation and prediction of new material structures and properties.[136]

The Transformer architecture has recently become a cornerstone in DL, revolutionizing various fields from natural language processing (NLP) to image generation. Initially designed for NLP, the Transformer model employs self-attention mechanisms to capture dependencies between data points, irrespective of their position in the sequence.[137] This versatile architecture has been adapted to multiple domains, giving rise to models like the vision Transformer (ViT),[138] generative pretrained Transformer (GPT),[139] and bidirectional encoder representations from Transformers (BERT).[140]

One of the key innovations of the Transformer is its self-attention mechanism, which allows it to process input data in parallel rather than sequentially, unlike earlier models such as RNNs and LSTM networks. The self-attention mechanism is computed as

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V; \quad Q = XW_{\mathrm{Q}}, K = XW_{\mathrm{K}}, V = XW_{\mathrm{V}} \tag{12}$$

where Q, K, and V are learned matrices for queries, keys, and values, respectively. This mechanism enables the model to focus on the most relevant parts of the input sequence regardless of their position. To further enhance its capabilities, the Transformer model incorporates multihead attention, which is calculated as

$$\mathrm{Multihead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_i)W_{\mathrm{O}} \tag{13}$$

where each head computes attention as

$$\mathrm{head}_i = \mathrm{Attention}(QW_i^{\mathrm{Q}}, QW_i^{\mathrm{K}}, VW_i^{\mathrm{V}}) \tag{14}$$

The model also uses positional encodings (PE) to account for word order:

$$\mathrm{PE}_{(\mathrm{pos},2i)} = \sin\left(\frac{\mathrm{pos}}{10000^{2i/d}}\right), \quad \mathrm{PE}_{(\mathrm{pos},2i+1)} = \cos\left(\frac{\mathrm{pos}}{10000^{2i/d}}\right) \tag{15}$$

In NLP, the Transformer architecture is the foundation for LLMs such as GPT and BERT. GPT is a unidirectional Transformer model that aims to autoregressively predict the next word in a sequence based on the previous words,[139] excelling in tasks like text generation. GPT has shown promising capability for parsing, searching, filtering, classifying, summarizing, and unifying materials data from scientific literature.[141] On the other hand, BERT is a bidirectional Transformer model that uses a masked language modeling objective to predict masked words by considering both the left and right contexts of a word in sentences.[140] The BERT model has been used to perform document classification, named entity recognition, and relation classification in the materials science domains.[142]

In image processing, the ViT applies the core Transformer architecture[138] by breaking down images into patches and treating them similarly to words in a sequence. Self-attention mechanisms are used to capture relationships across the entire image. ViT has demonstrated state-of-the-art results in various image classification tasks and is increasingly replacing CNNs in visual tasks. For instance, ViT has proven to be highly effective in materials classification based on XRD and FTIR spectra.[143]

The self-attention, multihead attention, and positional encodings Transformer models capture complex relationships between data points. As a result, Transformer-based models such as GPT, BERT, and ViT have surpassed RNNs, LSTMs, and traditional CNN in performance across various domains.

Materials Informatics. Material databases, descriptors, feature engineering, and QSPR are central to materials informatics. Material

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databases are the backbone of materials informatics, providing structured repositories of material properties derived from multiple sources, including experiments, first-principles calculations, simulations, and modeling.

Experimental data provide real-world measurements of material properties, representing the most accurate measure of how materials behave under various conditions. However, the experimental data is often limited due to the cost, time, and resources required to generate it.[124] Modern repositories blend laboratory results with HT computational outputs to enlarge these sparse measurements into richer training data sets. Databases, such as Citrine,[144] National Institute of Standards and Technology (NIST) materials data repository together with its Joint Automated Repository for Various Integrated Simulations (JARVIS) computational suite,[145] and National Institute for Materials Science (NIMS) materials data repository,[146] collectively host millions of curated entries that couple experimental records with simulation data. Complementing these public repositories, the enterprise-scale Granta MI platform centralizes the management of proprietary materials data, integrating experimental and simulation records across the product lifecycle to support engineering workflows and compliance.[147]

Density-functional theory (DFT) is, in principle, an exact quantum mechanical framework for any ground-state electronic system.[148] In practice, the exact exchange−correlation (XC) functional is unknown, so calculations rely on approximate functionals; these approximations can introduce systematic errors that make DFT predictions unreliable for specific materials or properties.[149] More sophisticated choices—such as PBEsol, the Perdew−Burke−Ernzerhof functional reparameterized for solids, or carefully tuned DFT + U, where U is the on-site Coulomb interaction added to capture strong electron−electron correlations that conventional functionals miss—usually improve accuracy but raise the computational cost, limiting HT studies.[150] To alleviate this cost, hybrid strategies combining DFT with ML techniques have shown promise as an effective way to enhance efficiency with comparable accuracy to that of DFT. Such ML surrogates match—but cannot surpass—the accuracy of the underlying DFT calculations while delivering speed-ups of up to 1,000 times in computational time, depending on the system size and the implementation of the model.[150,151]

Studies have demonstrated that integrating ML surrogates with DFT can accelerate the computational time significantly. A recent study showed a speed-up of 1,184 times using a hybrid DFT-ML approach,[152] while another ML-DFT framework was reported to be orders of magnitude faster than conventional DFT, with scalability to larger systems.[151] Additionally, the OrbNet Denali model achieved a speed-up of up to 3 orders of magnitude over standard DFT calculations.[153] These findings collectively indicate that ML surrogates can substantially accelerate the DFT calculations.

This acceleration is particularly significant given the vast data sets required for ML−DFT hybrid approaches. Large public materials databases have become a foundational infrastructure to support this data-intensive paradigm. The statistics and descriptions presented here are based on data from the Open Databases Integration for Materials Design (OPTIMADE) provider dashboard,[154] which serves as a unified access point to a wide range of computational and experimental repositories. Among the most prominent general-purpose platforms are Automatic-Flow (AFLOW)[155] and the Materials Project,[39] both supporting HT, DFT-based materials design: AFLOW hosts data for 3,530,330 structures, while the Materials Project provides DFT-computed properties for 154,387 structures. The Alexandria library[156] extends this scope with 5,635,291 stable and metastable structures, and the Computational Materials Repository (CMR)[157] offers 16,905 structures. The Novel Materials Discovery (NOMAD) repository[158] stands out with 12,116,021 entries. Complementing these are ML-focused resources such as the Matterverse,[159] which includes predictions for over 31 million hypothetical and known materials, and the Material-Property-Descriptor Database (MPDD)[160] offering 4,457,640 structures tailored for HT featurization and ML model deployment. The Open Materials Database (OMDB)[161] supports programmatic access to 68,566 structures optimized for computational workflows. Experimentally determined crystal structures are equally critical: the Crystallography Open Database (COD)[162] provides 525,560 entries for organic, inorganic, and metal−organic compounds (excluding biopolymers); the Theoretical COD (TCOD)[163] contains 7,488 theoretically refined or calculated structures; and the Open Database of Xtals (ODBX)[164] offers 522,847 *ab initio* predicted structures. Additionally, the Materials Platform for Data Science (MPDS)[165] compiles Pauling file data extracted from over 0.5 million publications. Federated platforms like Materials Cloud,[166] with 38,060 curated structures, further enrich the ecosystem with high-profile data sets, including those on ternary electrides, Li-ion conductors, and topological materials.

The landscape expanded sharply with the Open Materials 2024 (OMat24) release, which adds >100 million single-point and relaxation calculations for ∼3 million inorganic crystals.[167] Complementing the inorganic resources, the newly announced Open Molecules 2025 (OMol25) data set contributes ∼83 million unique molecular systems—small molecules, biomolecules, metal complexes, electrolytes—and provides baseline SE(3)-equivariant models for molecular property prediction.[168] Together, these data troves offer unprecedented coverage of both crystalline and molecular chemistries. Benchmarking infrastructure has kept pace: the Matbench and Matbench-Discovery leaderboards maintained by the Materials Project now comprise 13 tabular prediction tasks plus a discovery challenge, enabling transparent, continually updated comparisons of algorithmic progress across the community.[169,170]

Moreover, materials databases often include results from molecular dynamics (MD) simulations and finite element modeling (FEM). MD simulations, governed by Newton's laws of motion, are beneficial for understanding the dynamic behavior of materials at the atomic scale.[171] While MD can be performed using either classical interatomic potentials or first-principles methods such as DFT, classical MD provides a computationally efficient way to model dynamic processes that are challenging to capture with static DFT. Time-dependent DFT (TD-DFT) extends DFT to describe systems that are subject to time-dependent external fields. It allows the study of excited states as opposed to static DFT, which is limited to the ground state. Hence, TD-DFT coupled with MD enables a nonthermal, electronic excitation-driven pathway for ultrafast amorphization, establishing the Sc−Sb−Te alloy as a superior candidate for optical memory and computation due to its dual capability for fast SET and RESET operations.[172] Simulation tools like TDDFT-MD, Python-based Battery Mathematical Modeling (PyBaMM), and COMSOL Multiphysics enable high-fidelity, physics-based simulations at the continuum level, producing rich data sets for system-level analysis.[172−174]

Given the exponential growth of computational and experimental materials data, the need for cohesive data sharing has become increasingly critical, prompting a shift toward the standardization and centralization of materials databases. In response, several database providers collaborated to establish a universal Application Programming Interface (API) specification for seamless integration across the platforms. This effort began with the OPTIMADE workshop in 2016 and was refined through the annual CECAM workshop.[175,176] The resulting platform, OPTIMADE, now connects 24 providers and 29 subdatabases, with over 27 million structures. The stable release of version 1.0 enables broad access and compatibility, thereby advancing open materials discovery and development.

Despite the progress in unifying structured materials data through platforms like OPTIMADE, much valuable materials knowledge remains locked within unstructured formats—namely, scientific publications, patents, and technical reports. To fully leverage this wealth of information, it is essential to complement structured database efforts with tools that can systematically extract and organize insights from the scientific literature text. Therefore, the materials science community has also seen rapid advancements in NLP/BERT/LLM-enabled tools that extract and analyze materials information from scientific literature.[177−192] For instance, BatteryBERT and OpticalBERT—materials-domain-specific language models—are set

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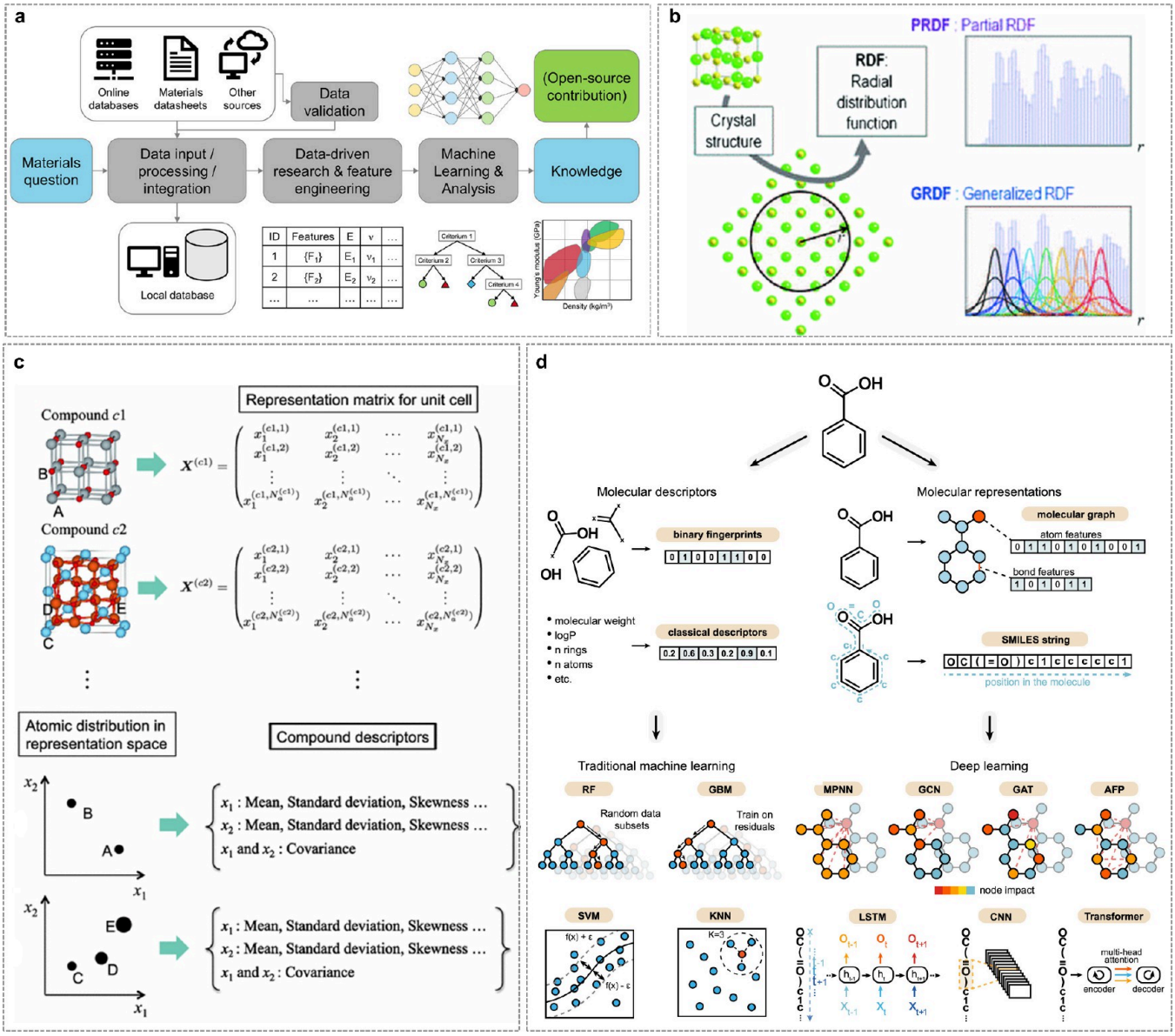


Figure 2. Overview of the Materials Informatics workflow. (a) A data-driven workflow for answering materials science questions and integrating data from online databases, materials datasheets, and local databases. Key steps include data validation, feature engineering, and ML construction. Reprinted with permission from ref 215. Copyright 2020 American Chemical Society. (b) Structural descriptors represent atomic interactions and provide insights into atomic arrangements in crystal structures. (c) Representation matrices for unit cells and atomic distributions. Figures (b) and (c) are adapted with permission under a Creative Commons CC BY 4.0 License from ref 216. Copyright 2018 The Author(s). (d) Molecular descriptors for predictive modeling. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 217. Copyright 2022 The Author(s).

to become available for the community to use in extracting insights into battery and optical materials, respectively, from the scientific literature.[193,194] The recently published PhotocatalysisBERT has been introduced, enabling researchers to leverage knowledge extraction from photocatalysts-related textual information.[195] These tools and platforms collectively enhance the accessibility, scalability, and reproducibility of materials science research, paving the way for accelerated innovation.

While literature mining tools help unlock insights from unstructured text, materials science data remain inherently heterogeneous, spanning structured databases, natural language descriptions, atomic structures, spectra, and images. This diversity poses challenges for holistic analysis and model generalization. To overcome this fragmentation, recent advances in multimodal deep neural networks (DNNs) are beginning to unify the heterogeneous data landscape of materials science.[196] Early graph models such as MatErials Graph Network (MEGNet) and Atomistic Line Graph Neural Network (ALIGNN) performed well with atomic-graph inputs.[197,198] The same for Materials Exploration (GNoME), which was trained on more than 30 million DFT structures and predicted ≈380,000 previously unknown stable crystals that have since been experimentally verified.[199] However, within a single architecture, the newest 'foundation' models accept multiple representations—crystal graphs, Simplified Molecular Input Line Entry System (SMILES) strings, 3-D coordinates, natural-language text, spectra, and microscopy images.[200] The Materials 3-body Graph Network (M3GNet) integrates data of different quantum-chemical fidelities through multifidelity training.[159] IBM's Foundation Models for Materials (FM4M) and the Multimodal Learning for Materials (MultiMat) framework align 1-D, 2-D, and 3-D descriptors via contrastive learning.[196,201] Language-centric models like MatGPT couple crystal graphs with large-scale text corpora, enabling structure-aware question answering and retrieval.[202] By

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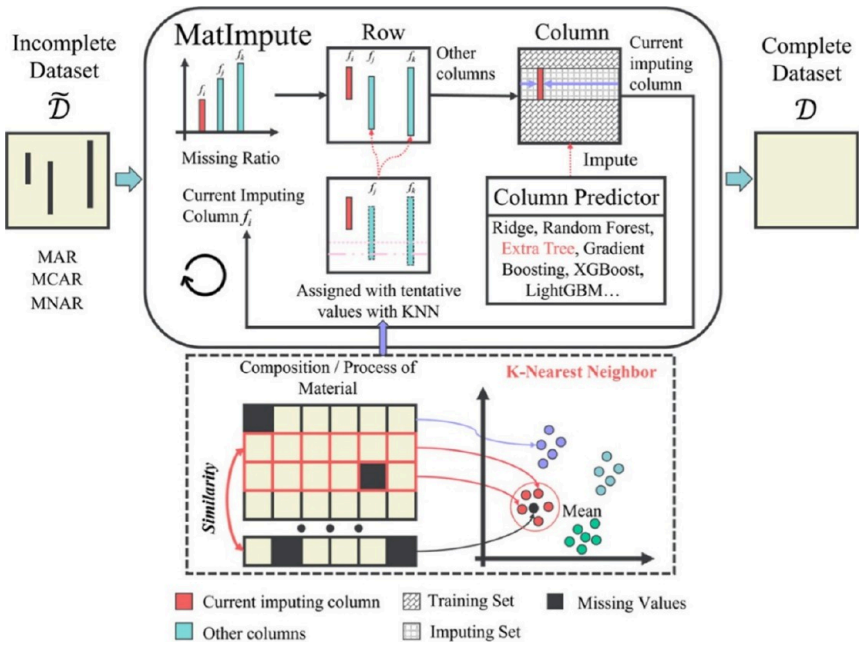


Figure 3. Diagram illustrates the MatImpute framework, where missing data in materials science data sets are iteratively imputed by KNN for tentative value assignment and column predictors to refine predictions, resulting in a complete data set. Reprinted with permission from ref 218. Copyright 2025 American Chemical Society.

learning transferable latent spaces, these multimodal models typically achieve 10−30% accuracy gains on small, domain-specific data sets, support cross-modal queries (*e.g.,* 'find a crystal whose XRD pattern matches this spectrum'), and provide a standard backbone for property prediction, inverse design, and autonomous experimentation.[199]

Material descriptors are numerical or categorical representations of materials data that serve as inputs to learning models. Material descriptors can be broadly categorized into several types. Composition-based descriptors capture the elemental composition of the material, such as atomic fractions, electronegativity, atomic radii, ionization potentials, elemental properties, bond fractions, and chemical ordering.[129,203,204] Structure-based descriptors describe the atomic arrangement in the crystal lattice, such as bond lengths, coordination numbers, crystal lattice parameters, radial distribution function (RDF), partial radial distribution function (PRDF), packing efficiency, symmetry descriptors, orbital field matrix, Coulomb matrix, sine Coulomb matrix, and a bag of bonds.[129,203−207] Electronic descriptors involve properties related to the electronic structure of the material, such as E_g, charge densities, density of states (DOS), Fermi level, highest occupied molecular orbital (HOMO), and lowest unoccupied molecular orbital (LUMO).[203,207,208] For certain quantum-materials applications—most prominently the discovery of topological insulators and semimetals—the DOS must be analyzed in tandem with symmetry-resolved structural descriptors (*e.g.,* space group, Wyckoff positions, site-symmetry irreducible representations). Only this combined electronic−structural view captures the band inversions and symmetry indicators that define topological invariants, whereas the DOS spectrum alone is often inconclusive.[209,210] Mathematically, material descriptors are represented as feature vectors X represented as

$$X = [x_1, x_2, ..., x_n] \tag{16}$$

where each x_i represents a specific property or feature.

Practical feature engineering is essential for the success of ML models. Techniques like PCA help reduce the dimensionality of feature vectors by identifying the most critical descriptors while preserving the maximum variance in the data.[211] Since material descriptors often span different scales, feature scaling ensures equal contribution from all descriptors to the model's predictions. This is commonly done by rescaling each descriptor to a standard range, such as 0 to 1,[212] improving model performance and stability.

Lastly, QSPR forms the core of predictive modeling in materials informatics.[213,214] QSPR models establish mathematical relationships between the structure of a material, described by its descriptors, and its properties. These models allow researchers to predict material properties based on their structural, compositional, or electronic features without costly experiments or simulations.

Figure 2 illustrates the process of generating material descriptors, feature engineering, and predictive modeling. Figure 2a presents the overall flow of materials informatics, from data input and integration to ML and knowledge generation.[215] It shows how researchers work with local and online databases, applying feature engineering and ML techniques to answer questions about specific materials. Figure 2b delves into structural descriptors, demonstrating how crystal structures are represented by RDF, including both PRDF and generalized RDF (GRDF), which capture atomic interactions over distances. Figure 2c highlights the creation of representation matrices for unit cells, atomic distributions, and compound descriptors, emphasizing how these representations enable a better understanding of material behavior.[216] Lastly, Figure 2d displays molecular descriptors, such as SMILES strings, which are crucial in predicting the properties of molecules using ML models.[217] Together, these elements enable precise predictive modeling in materials science.

Dealing with Missing Material Data. In materials science, even a modest missing data ratio of 0.15 can significantly impair the performance of predictive models.[218] Imputation, the statistical remedy of substituting every missing entry with one or more plausible values sampled or estimated from the joint distribution of the observed data,[219] enables subsequent analyses to proceed on 'completed' data sets without discarding records. Single imputation inserts one deterministic estimate (for example, a column mean or a k-nearest-neighbor vote), whereas multiple imputation draws $m \geq 2$ stochastic replacements—where m is simply the number of independently imputed 'complete' data sets created—so that downstream uncertainty can be propagated by pooling the results across those m data sets.[220] Crucially, the reliability of any imputation depends on the missing-data mechanism—missing completely at random (MCAR), missing at random (MAR), or missing not at random (MNAR)—and on whether the model used to predict the gaps captures the multivariate correlations present in the full

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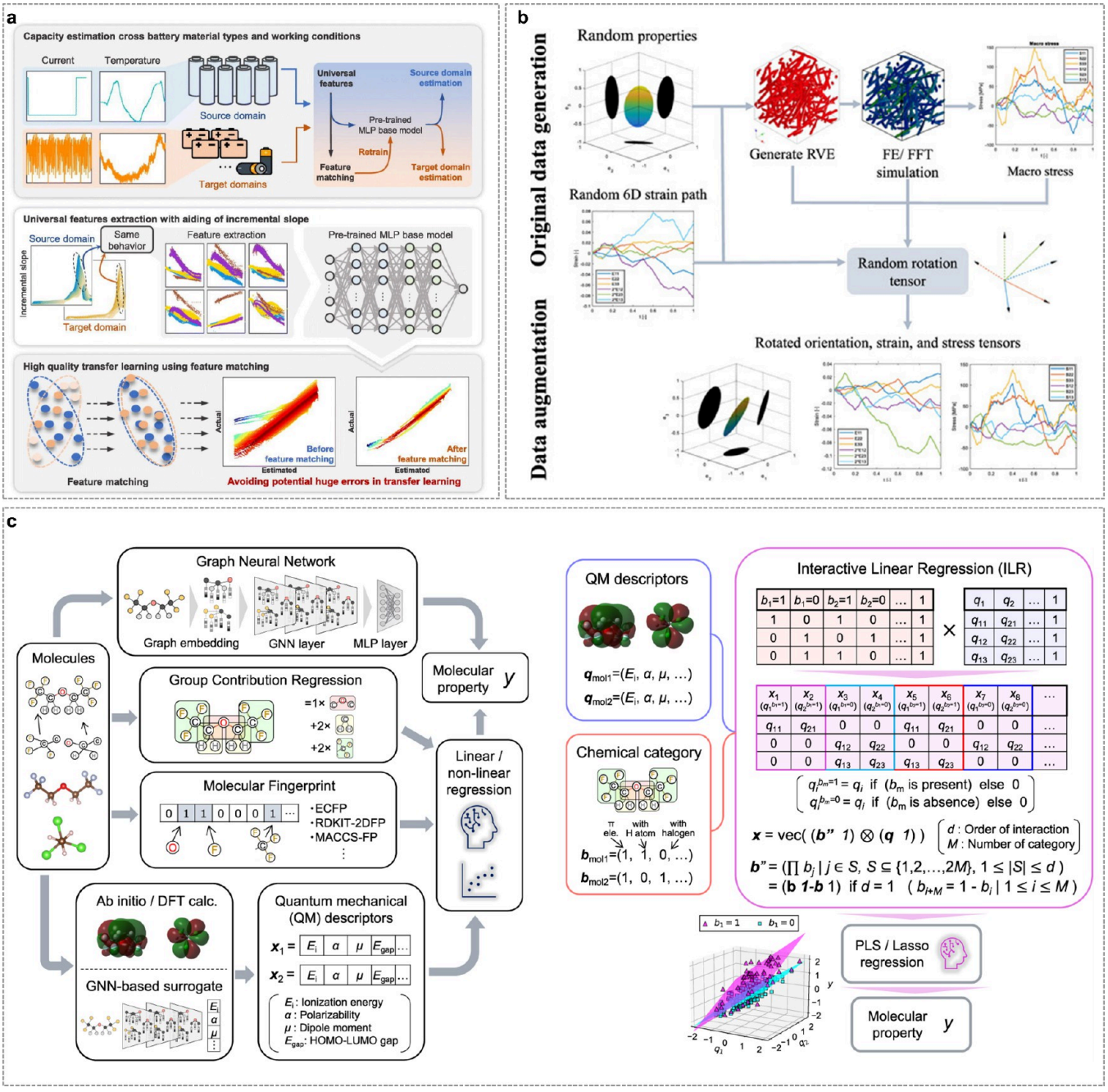


Figure 4. AI techniques for enhancing model performance on small materials data sets. (a) TL was applied to battery capacity estimation across different battery types and conditions using feature extraction and feature matching. Reprinted with permission from ref 229. Copyright 2023 Elsevier Ltd. (b) Data augmentation through random rotation of strain and stress tensors in materials simulations. Adapted with permission under a Creative Commons CC BY 4.0 License from ref 231. Copyright 2024 The Author(s). Published by Elsevier Ltd. (c) Ensemble learning combining molecular descriptors and GNNs for molecular property prediction. Adapted with permission under a Creative Commons CC BY 4.0 License from ref 232. Copyright 2024 The Author(s).

population.[221] By choosing the optimum missing-data mechanism, imputation can effectively handle missing data across diverse data sets and enhance ML predictions.[222] By improving data integrity and preserving essential statistical characteristics, imputation ensures reliable predictions without compromising data set size.[76,223]

Several imputation methods have been utilized to deal with the missing data in the materials science domain. For example, the K-Nearest Neighbor (KNN) combined with DNN improved the prediction of material properties, achieving a coefficient of determination (R^2) score of 0.973.[223] Similarly, multivariate imputation by chained equations (MICE) has been used to impute the missing data of the synthesis condition data set of lithium-ion batteries, enabling the ML model to achieve high predictive accuracy.[76] Another study coupled imputation with Monte Carlo flow-based generative model (MCFlow) to address missing data in materials science databases, achieving robust imputation and improved regression performance.[224] Meanwhile, the MultiScaleCNN-Transformer model with particle swarm optimization (PSO)[225] was developed to impute missing acoustic absorption coefficients for transversely isotropic porous materials. For partial mask imputation, the decoder utilized both feature data from the encoder and known coefficients of adjacent frequencies, leading to high accuracy and stable performance across the entire test set.[226] Lastly, the MatImpute, as shown in Figure 3, combines KNN and

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Extra Trees (ET) to iteratively impute missing data by leveraging row-wise similarities and predictive modeling.[218] Columns are imputed in ascending order of missing ratios, with tentative values assigned using KNN and refined through ET-based predictions, iteratively updating until all data are imputed. This approach minimizes imputation-induced errors (IIE), outperforming other methods in reducing root-mean-square error (RMSE) and Wasserstein Distance (WD), and preserving data correlations across various missing ratios and types.

AI Techniques for Small Size Material Data Sets. ML and DL models typically perform best when trained on large data sets, allowing them to capture complex patterns and relationships within the data.[215,227] However, in materials science, acquiring vast amounts of labeled data can be difficult due to the experimental or computational cost associated with generating high-quality data sets.[124] This poses a challenge for ML models, as small data sets often lead to poor generalization, limiting the models' ability to make accurate predictions.[124] TL is a powerful approach that integrates the strengths of both supervised and unsupervised learning. It is particularly revolutionary in supervised learning as it enables the reuse of knowledge from pretrained models, reducing the need for extensive labeled data. TL can significantly lower the cost of data annotation while building precise structure−property relationships, making it highly promising for applications in molecular and materials science. By leveraging existing knowledge, TL not only improves prediction accuracy but also minimizes the amount of new data required for training, enhancing efficiency and scalability.[228]

For example, TL has been effectively applied to battery capacity estimation in materials science, as demonstrated in a study on lithium-ion batteries.[229] The researchers employed a feature-matching-based TL (FM-TL) method, which adapted capacity estimation models across different battery types and working conditions, as shown in Figure 4a. Using TL, the work showed significant improvements in prediction accuracy with reduced data requirements.

Another powerful technique to enhance ML models, especially when working with small data sets, is data augmentation. In materials science, data augmentation can artificially expand the size of a data set by creating modified versions of existing data points.[128,230] This method essentially increases the diversity of the training data without collecting new samples. Figure 4b illustrates how random orientations and rotations of stress tensors and strain paths are used in data augmentation, creating a more extensive and diverse data set for better predictive performance.[231]

Ensemble learning is a highly effective approach, particularly for improving the generalization of ML models trained on small data sets. Ensemble learning involves combining predictions from multiple models to reduce the risk of overfitting and enhancing predictive performance. For example, ensemble learning methods have shown substantial improvements in both interpolation and extrapolation tasks in molecular property predictions, such as estimating boiling points and solubility.[232] Ensemble learning can effectively mitigate the biases present in small data sets by aggregating predictions from different models, leading to more robust prediction accuracy. Figure 4c highlights an ensemble learning case, where molecular descriptors and GNNs are used to predict various molecular properties through various regression techniques.

Finally, LLMs are also being explored to enhance small data set analysis in materials science by enabling interactive and context-based predictions. LLMs can generate insights by understanding and processing materials-related text and structured queries, making them highly adaptable tools for materials discovery. By generating and evaluating candidate structures from simple user prompts, they further demonstrate a powerful capability to augment human expertise in materials design and analysis.[136] Nevertheless, synthetic or augmented data produced by these models come with notable risks: generative systems can hallucinate chemically implausible structures or properties,[233] and repeated recycling of such outputs may trigger model collapse, whereby the learned distribution drifts toward low-diversity artifacts and away from the true materials manifold.[234] Moreover, state-of-the-art graph neural networks—even when heavily augmented—show marked accuracy drops on deliberately out-of-distribution (OOD) generations, underscoring the need for rigorous domain checks and uncertainty quantification.[235] Careful curation and validation of augmented data are, therefore, essential components of any responsible training pipeline.

RECENT APPLICATIONS OF AI FOR MATERIALS DISCOVERY

Materials discovery is the process of identifying new materials with targeted properties through design, synthesis, and performance/property evaluation. Traditionally reliant on trial-and-error experimentation and first-principles calculations, the field has evolved with the advent of data-driven science, HT screening, and automation, making the process more efficient and predictive. Initiatives like the Materials Project utilize DFT databases to guide materials selection,[39] while ML models enable property prediction and inverse design, bypassing conventional bottlenecks.[124] This section explores three key AI-driven areas transforming the field: materials structure generation and identification, materials property prediction, and HT screening and computational materials design. Collectively, these advances demonstrate AI's transformative potential in accelerating discovery and expanding the frontier of materials science beyond what was previously feasible using conventional approaches alone.

Materials Structure Generation and Identification. Recent advancements in generative models have introduced powerful tools for structure prediction in materials science,[236,237] significantly enhancing the ability to design and identify complex materials structures. Well-established evolutionary algorithms, most notably Universal Structure Predictor: Evolutionary Xtallography (USPEX)[238] and Crystal structure AnaLYsis by Particle Swarm Optimization (CALYPSO),[239] remain the state of the art for unconstrained *ab initio* searches and still serve as essential baselines. In fact, generative models are already transforming the field. They enable the design of crystal structures with explicit symmetry constraints,[240] lattice-constrained perovskite materials,[241] and metamaterial systems.[242] They have also advanced the identification of materials structures, such as predicting relaxed structures from their unrelaxed counterparts,[243] significantly reducing reliance on traditional trial-and-error methods.

One prominent example is the Crystalyze framework, developed by Riesel et al., which applies generative ML models to predict crystal structures from powder diffraction patterns.[244] Traditional methods for structure determination, such as trial-and-error or expert-driven techniques, are time-consuming and resource-intensive. In contrast, Crystalyze leverages CNNs and multilayer perceptron to encode diffraction patterns into a latent space, significantly improving the speed and accuracy of the structure predictions (Figure 5a). The model has been particularly effective in identifying novel high-pressure compounds through powder XRD (PXRD) pattern analysis, demonstrating its potential to accelerate materials discovery. Pushing beyond structure determination toward fully *de novo* design (*i.e.*, generating crystal structures entirely from scratch, without any experimental template or prototype), Microsoft recently released MatterGen, a diffusion-based framework that treats a periodic solid as a joint point cloud of atom types, fractional coordinates, and lattice vectors.[245] During training, Gaussian noise is added simultaneously to these discrete−continuous variables, and a special Euclidean group in three dimensions (SE(3))-equivariant GNN learns to reverse the noise by iteratively denoising the crystal back to a low-energy state, while a learned formation-energy surrogate biases the trajectory toward thermodynamic stability. At inference, repeated denoising steps refine chemical identities, atomic positions, and unit cell parameters, yielding inorganic crystals that are both novel and energetically favorable.

Another generative model making strides in structure prediction is a Generative Learning to Inform the Design of Electrodes (GLIDER), an AI framework designed to optimize the multiscale nanostructures of hydrogen fuel cell catalyst layers (CLs).[246] Developed by Niu et al., GLIDER integrates vector quantized-VAE (VQ-VAE) and CNNs to reduce the dimensionality of high-complexity CL designs, such as

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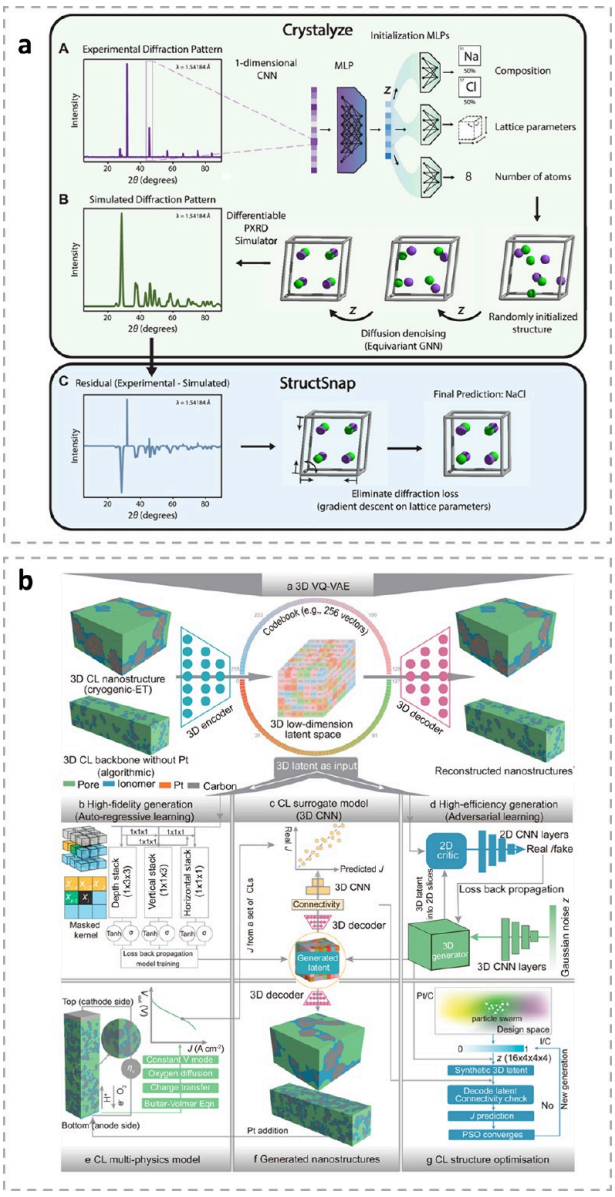


Figure 5. Generative models for materials structure prediction. (a) The Crystalyze framework uses CNNs to predict crystal structures from powder diffraction patterns. It efficiently reconstructs crystal lattices by encoding diffraction data into a latent space and matching the simulated patterns with experimental ones. Reprinted with permission from ref 244. Copyright 2024 American Chemical Society. (b) GLIDER model for optimizing 3D nanostructures in hydrogen fuel cell catalyst layers using VQ-VAEs and CNNs. It compresses high-complexity CL designs, such as Pt/C systems, into a latent space for improved structure-performance relationships. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 246. Copyright 2024 The Author(s).

platinum−carbon and ionomer systems.[246] GLIDER enables efficient 3D CL structure design based on electrochemical performance data by compressing these nanostructures into a low-dimensional latent space. As illustrated in Figure 5b, the architecture of GLIDER demonstrates how the framework encodes and decodes these intricate 3D structures. The model achieved significant performance improvements, with a high correlation between the optimized CL designs and experimental data.

The field of generative classifiers has also seen notable advancements, particularly in the automation of phase diagram mapping. Arnold et al. developed Bayesian generative models that exploit statistical mechanics principles to classify phase transitions without significant human intervention.[247] This approach improves the accuracy of phase classification and offers a more natural method for detecting phase boundaries. Generative classifiers model the conditional probability of observing data for a given phase label, while discriminative classifiers directly predict phase labels based on the data. This difference allows for generative models to use the physical principles governing the system. The generative approach quantitatively reproduces the phase boundaries by calculating a scalar indicator of phase transitions, aligning with well-known analytical solutions. This demonstrates the robustness of the generative framework in capturing complex physical phenomena. Similarly to how AlphaFold, by using large-scale data sets, achieved breakthroughs in protein structure prediction, generative models in materials science can exploit high-quality data sets to improve predictive modeling for phase boundaries and other complex materials properties.[248]

Complementing these generative methods, Dai and Glotzer introduced an active-learning framework that couples Gaussian-process regression with information-based acquisition functions to adaptively sample state points.[249] Their scheme maps boundaries using barely 20% of the simulations required by conventional grid searches while maintaining comparable precision, as shown for active-passive colloid mixtures and quasicrystal transitions. Miryashkin et al. extended the probabilistic paradigm by combining Gaussian-process regression with MLPs to infer full composition−temperature phase diagrams (e.g., K−Na), providing uncertainty quantification and an active-learning strategy for optimal data acquisition.[250]

Regarding phonon spectra prediction, the virtual node GNN (VGNN) developed by Okabe et al. offers another generative approach incorporating virtual nodes into standard GNNs.[251] The VGNN framework predicts phonon band structures directly from atomic coordinates, providing a flexible and efficient alternative to costly density functional perturbation theory calculations. Virtual nodes are added to the crystal graph, allowing for flexible output dimensions and prediction of phonon spectra across various materials. Okabe et al. first benchmarked VGNN predictions against an HT density functional perturbation theory (DFPT) database, then cross-checked selected cases with inelastic neutron-scattering phonon spectra, showing close agreement.[251] Thus, while the large-scale validation is computational, the study also provides targeted experimental corroboration of the model's accuracy. The models were trained using an *ab initio* DFPT computational database for phonon dispersion in the harmonic model.[252] This data set includes material structures, the same as the primitive structure obtained from the Materials Project,[39] second-order derivatives of energies with respect to atomic perturbations, and phonon dispersion along symmetric paths of 1,521 crystalline inorganic materials. The materials, ranging from 2 to 40 atoms per unit cell (with an average of 7.38), were used with only the phonon dispersion along highly symmetric paths for training.

Generative models have shown remarkable capabilities for materials microstructure generation, particularly with the introduction of denoising diffusion probabilistic models (DDPMs).[253] These models have been highly effective in generating high-fidelity 2D and 3D microstructures, such as inclusion and spinodal decomposition materials, by learning the probability distribution of raw data. DDPMs also support conditional generation, enabling the design of microstructures with specific permeability ranges and validating these designs using techniques like the lattice Boltzmann method.[254] This study integrated supplementary feature encoding embeddings with time feature encoding into the diffusion model, allowing precise control over various physical attributes,[253] as shown in Figure 6a.

Hybrid methods that combine DDPM with GANs have further enhanced the quality of generated microstructures, producing accurate 3D volumetric data from 2D images while improving computational efficiency.[255] Another approach integrates multiscale features of U-net with GANs, using hierarchical feature aggregation and multiscale channel aggregation to enhance feature extraction, along with a convolutional block attention module for improved

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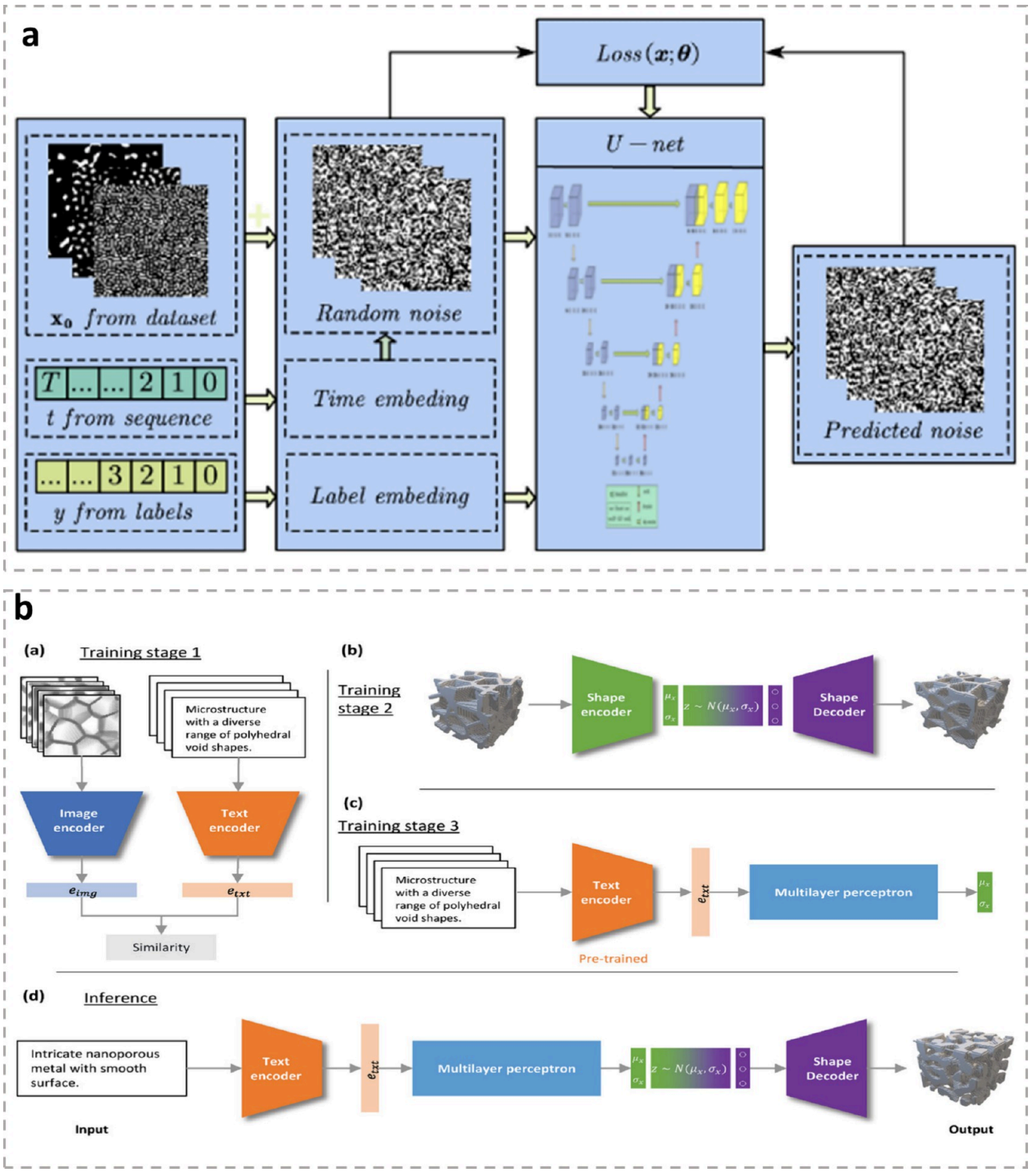


Figure 6. Generative models for materials structure generation. (a) DDPMs for microstructure generation can produce high-fidelity 2D and 3D microstructures by learning the probability distribution of the data. The models allow conditional generation, enabling the targeted design of microstructures with specific physical properties. Note: The word 'embeding' in the figure contains a typographical error and should read 'embedding.' Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 254. Copyright 2024 The Author(s). (b) A hybrid U-net and GAN model for reconstructing 3D materials structures, such as lithium battery NCM Oxide cathodes, using multiscale feature aggregation and attention mechanisms to enhance accuracy while reducing computational costs. Reprinted with permission under a Creative Commons CC BY 4.0 from ref 258. Copyright 2024 The Author(s).

computational efficiency. This model achieved high fidelity in reconstructing 3D structures like NCM Oxide cathodes, outperforming traditional GANs in accuracy while reducing computational costs efficiency.[256] Furthermore, diffusion-based generative models (DGMs) have been proposed for designing multifunctional composite microstructures.[257] This framework integrates microstructure characterization, property analysis, and inverse design using conditional DGMs, supported by CNN-based surrogate models, to predict and generate microstructures with specific properties. Trained in microcomputed tomography images, this model

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demonstrated robustness, producing high-fidelity samples that meet design targets such as volume fraction, elastic modulus, and light intensity while avoiding mode collapse and reducing computational costs.[257]

Another notable approach is the text-to-microstructure deep generative network, which uses text prompts to generate 3D microstructures, offering flexibility in design without needing additional optimization.[258] As shown in Figure 6b, the approach can help mitigate the risk of overfitting, save computational resources, and improve generalization.[258]

Materials Property Prediction. Materials property prediction can be defined as the process of using models to estimate the properties of a material based on its structure, composition, or processing history.[259−261] Recent ML frameworks have significantly enhanced the prediction of various materials properties across various applications in materials science. These models leverage the predictions of mechanical,[262−264] electrochemical,[265] magnetic,[266] rheological,[267] viscoelastic,[268] catalytic,[269] dielectric,[270] optical,[271] acoustic,[272] electrical,[273] surficial,[274] thermal,[275] biochemical,[276] physicochemical,[277] tribological,[278] hydrodynamic,[279] nuclear and radiological,[280,281] thermodynamic,[282] quantum,[283] crystallographic and molecular,[284,285] and toxicological[286] properties of materials, reducing the need for exhaustive experimental testing.

One of the most promising frameworks is the gradient-boosted feature selection (GBFS) method, introduced by Jung et al., which is used for predicting the critical temperature (T_c) of superconductors.[287] This model integrates gradient boosting with feature selection to reduce multicollinearity, optimizing the predictive accuracy of T_c. Trained on a data set of over 16,000 compounds, the model achieved an impressive R^2 value of 0.945 and a mean absolute error (MAE) of 3.54 K. The workflow of GBFS, shown in Figure 7a, highlights how iterative feature selection improves the prediction of T_c, making it highly suitable for complex materials like superconductors.

In molecular systems, predicting properties such as force fields and quantum chemical characteristics requires overcoming the challenge of biased molecular distributions in pretraining.[288] The fractional denoising (Frad) framework tackles this issue by decoupling noise design from physical constraints, allowing for more accurate property predictions.[289] Frad outperforms other models in predicting forces across various data sets, making it a powerful tool for accurately modeling molecular systems.[288]

For polymer properties, including glass transition temperature (T_g), E_g, and atomization energy, Polymer properties via the power of Natural language and Chemical language (PolyNC) framework integrates NLP with SMILES.[290] This multidomain strategy markedly outperforms traditional descriptor-based models in predicting polymer properties. PolyNC's transformer-based architecture simultaneously ingests chemical and textual information to forecast multiple properties at once, highlighting the power of multimodal data fusion for more accurate and reliable polymer property predictions.

In the context of catalytic activity, Zhou et al. developed an ML to predict the overall activity of Pt−Ru alloy catalysts in the hydrogen evolution reaction (HER).[291] By combination of DFT with a Crystal Graph Convolutional Neural Network (CGCNN), the framework predicts the catalytic performance of alloy surfaces by identifying active sites. The key innovation is the introduction of the equivalent site ratio (ESR) metric, which evaluates the distribution and effectiveness of active sites within the alloy. Figure 7b shows how the model predicts HER activity based on Ru content, achieving optimal performance at 20−30% Ru, validated by experimental data.[291] This method reduces the computational burden of DFT calculations and provides accurate predictions of the catalytic performance, especially for Pt−Ru alloys.

In the case of semiconductor defect prediction, traditional DFT is integrated with GNNs to estimate defect formation energies (DFEs) in materials like group IV, III−V, and II−VI Zincblende (ZnS) semiconductors.[292] GNN models such as CGCNN, MEGNet, and ALIGNN accelerate the prediction of native defects and impurities while maintaining high accuracy. As shown in Figure 7c, ALIGNN achieved the lowest RMSE of ∼0.15 eV, significantly reducing the

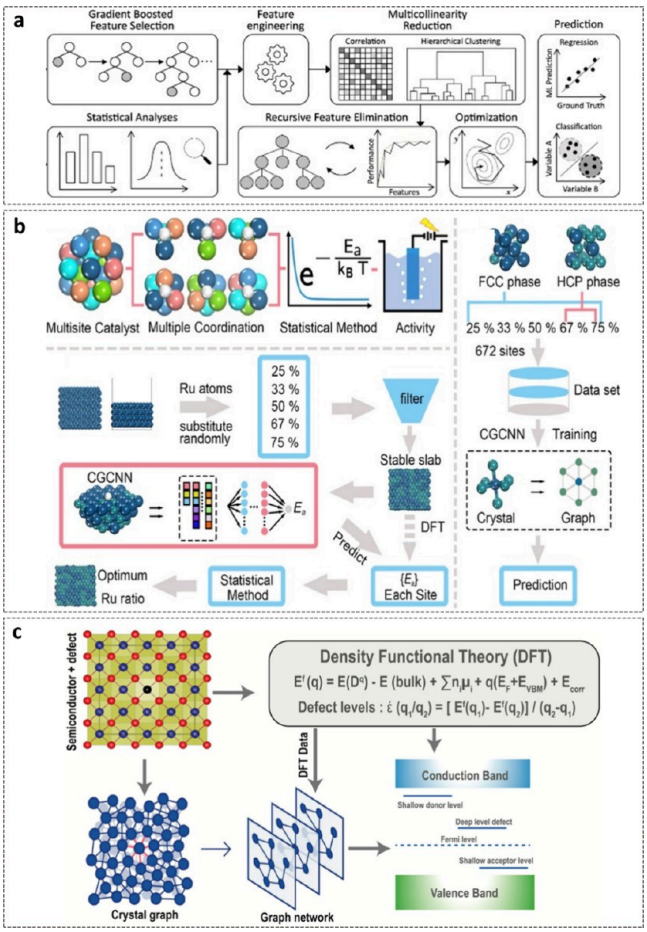


Figure 7. Overview of ML frameworks for materials property prediction. (a) The GBFS framework predicts superconducting T_c by integrating feature engineering, recursive feature elimination, and multicollinearity reduction. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 287. Copyright 2024 The Author(s). (b) ML-assisted framework for predicting catalytic activity in Pt−Ru alloy catalysts for HER, integrating DFT and CGCNN models. Adapted with permission from ref 291. Copyright 2024 American Chemical Society. (c) Prediction of DFEs in semiconductors using GNNs combined with DFT data. ALIGNN achieves the highest accuracy, significantly reducing computational costs for defect energy predictions across the semiconductor materials. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 292. Copyright 2024 The Author(s).

computational cost associated with DFT-based defect predictions.[292] This method marks a significant breakthrough in efficiently identifying defects in technologically critical semiconductors. Finally, the prediction of electrical conductivity and gas sensing performance (ammonia, toluene, benzene) in PANI/graphene (PANI/Gr) nanocomposites is achieved using artificial neural networks (ANNs).[293] The model, trained on a comprehensive data set, achieved high predictive accuracy with R^2 values of 0.984 for electrical conductivity and 0.967−0.983 for gas sensing responses. Derived σ-profiles, *i.e.*, probability distributions of screened surface charge densities that quantify the electrostatic and hydrogen bonding character of a molecule's surface,[294] offer more profound insights into molecular interactions within the nanocomposite, enabling targeted materials development with reduced experimental workload.

HT Screening and Computational Materials Design. HT screening in materials science refers to the automated or semi-automated process of rapidly computing or experimentally measuring the properties of a large number of materials.[23,295−306] This method is

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often used to identify promising candidates in vast chemical or structural spaces for specific applications. Meanwhile, computational materials design refers to the rational design of new materials using computational tools.[23,307−309] It focuses on understanding and predicting how structure, composition, and processing affect properties, and using this knowledge to design materials with desired functions. HT screening is often a tool or the first step within a broader computational materials design pipeline. HT methods can quickly scan candidates, while computational design methods explore why a material performs well and how to improve or tailor it.

HT screening and computational materials design rely on a two-tier architecture. DFT constitutes the physics-based tier, furnishing formation energies, band gaps, elastic constants, and ion-migration barriers with first-principles reliability. Its cubic scaling in electron count, however, makes brute-force enumeration of the $\gtrsim 10^6$ conceivable inorganic crystals computationally and energetically prohibitive.[310] The data now generated by DFT calculations have seeded a second, acceleration tier based on ML models. GNN models such as M3GNet reproduce DFT formation energies to $\leq$ 40 meV/atom while evaluating 10^5 compounds per second on a single graphics processing unit (GPU),[311] and large-scale frameworks like DeepMind's GNoME have used active learning to map 2.2 million low-energy crystals—an order-of-magnitude expansion of the known stable-materials space.[199] When coupled with autonomous synthesis platforms such as, autonomous laboratory (A-Lab), ML surrogates guide experiments in closed-loop fashion, converting month-long screening campaigns into week-scale discovery cycle.[312] Consequently, contemporary HT pipelines blend DFT for targeted high-fidelity validation with ML for rapid prescreening, adaptive sampling, and inverse design.

DFT as the Foundational Method for HT Screening and Computational Materials Design. DFT-based HT computational screening offers a powerful and efficient approach to materials design, enabling the identification of promising materials without requiring initial synthesis.[313−316] This method has been successfully used to discover new materials for applications, such as Li−air batteries and Li-ion batteries (LiBs),[317,318] and light-absorbing materials.[319] The screening and discovery process is depicted as a funnel plot in Figure 8, illustrating the generation, storage, characterization, and selection of suitable materials from structure−property data sets.[320] DFT being the core of first-principles materials modeling, in its limit of semilocal approximation, delivers reliable ground-state geometries, formation energies, interatomic forces, elastic tensors, and even phonon spectra for weakly correlated, closed-shell compounds—capabilities that underpin today's HT stability rankings.[23,39,321]

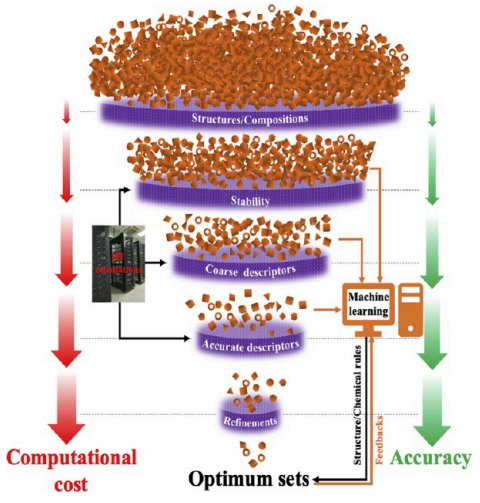


Figure 8. Funnel-type model of HT computational screening. Reprinted with permission from ref 316. Copyright 2020 Wiley Periodicals LLC.

The same approximations, however, limit DFT's reach. Standard functionals struggle with strong static or long-range correlation (Mott and charge-transfer insulators, 4*f*/5f oxides,[322] dispersion-dominated molecular and van-der-Waals solids,[323] polaronic self-trapping[323] and finite-temperature excited-state properties).[324] Post-DFT corrections—DFT+U, hybrid functionals, GW, dynamical mean-field theory, or correlated wave function methods—partially remedy these shortcomings but raise the computational cost by one to 3 orders of magnitude.

Because of this trade-off, the choice of exchange−correlation functional becomes pivotal. GGAs such as PBE and RPBE prioritize speed and exploit error cancellation; meta-GGAs (SCAN, r^2SCAN) offer better thermochemistry with only modest overhead; while screened and range-separated hybrids (HSE06, PBE0, ωB97X-V) yield more accurate band gaps and defect levels yet scale quadratically with system size.[325−327]

In the HT screening, the process begins with a range of potential structures and compositions, which are sequentially filtered through multiple stages including stability evaluation, coarse descriptors, accurate descriptors, and refinements. At each stage, the number of candidates is reduced, while the accuracy of the results improves. ML is integrated into the process to refine the outcomes by applying learned structure and chemical rules as feedback, allowing for faster, more accurate predictions. The trade-off between computational cost and accuracy is demonstrated with the model guiding the selection of optimal materials sets that effectively balance these factors.

Various quantum physical and chemical descriptors, including formation enthalpy, decomposition enthalpy, exciton binding energy, and E_g, are calculated using DFT. The HT screening funnel filters the extensive data set to identify optimal materials with desired properties, although some will still require future experimental validation.[328] The general workflow of HT computational materials screening and discovery is divided into five key steps,[316,329−333] outlined as follows:

The first step involves identification of the target structures. In this step, potential structures are queried from large data sets. This data set can be accessed directly from existing structure databases, including the Materials Project (MP), Inorganic Crystal Structure Database (ICSD), and Cambridge Structural Database (CSD), among others. New materials that are not present in these databases can also be explored using structure prediction techniques or elemental substitution based on known structure prototypes.

The second step involves thermodynamic stability screening of the structures. It involves electronic structure calculations that meet a specific level of accuracy. In some cases, dynamic simulations are also utilized for assessing the thermodynamic stability. The evaluation of a material's thermodynamic stability typically employs convex hull rules and phonon spectra.[23,334]

In the third step, preliminary descriptor screening is performed. This involves calculating initial quantum descriptors using low-cost DFT-based computational methods with a simpler exchange-correlation functional, *i.e.,* local-density approximation (LDA)[335] and generalized gradient approximations (GGA).[336] For example, GGA-calculated band gaps are often used as initial descriptors in screening optoelectronic semiconductors.[337]

In the fourth step, an accurate descriptor screening is performed. More precise computational techniques, like the HSE06 functional[338] and GW (derived from Green's function G and screened Coulomb interaction W)[339] methods improve descriptor accuracy, achieving better theoretical and experimental alignment.

In HT DFT screening, a large number of candidate materials are computationally evaluated for their physical and chemical properties by using quantum mechanical simulations. However, not all computationally promising materials are practical for use in real-world applications. That is where non-DFT filters come into play. Additional factors such as toxicity, elemental abundance, commercial value, and synthesis cost are used as final filters in the fifth step, or they can be used to filter the data as the initial step, followed by the standard screening procedure mentioned.

Arranging a virtual screening pipeline in a particular order offers significant benefits by balancing computational cost and predictive

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Table 1. Materials Open-Source Tools for HT Calculations Adapted from Ref 316

program	description	supporting packages
AFLOW[155]	an automatic framework for HT materials discovery, offering tools for DFT calculation preparation and analysis.	VASP, QE
AiiDA[348]	Automated Interactive Infrastructure and Database for Computational Science, designed to manage and share computational workflows and data.	QE
AMP2[349]	package for automated ab initio modeling of material properties	VASP
ASE[350]	set of tools and Python modules for setting up, managing, and analyzing atomic simulations.	VASP, CASTEP, ABINIT, QE, Gaussian,[351] *etc.*
Atomate[352]	software containing prebuilt workflows for computational materials science to compute and analyze material properties.	VASP, BoltzTraP[353]
COSMO[354]	a laboratory for developing computational tools for various scientific applications.	QE
Custodian[355]	a simple, robust framework for managing and monitoring VASP jobs, ensuring proper execution and error handling.	VASP, NwChem,[356] Qchem[357]
FireWorks[358]	a workflow management system for HT computation, which works on parallel server.	VASP
JARVIS	repository for integrating various simulations for materials science research.	VASP, QE, WIEN2k, BoltzTraP, LAMMPS[359]
Jasp[360]	Python module for creating VASP calculation scripts and automating job submissions.	VASP
JUPM2[361]	Python package for HT materials design focusing on functional semiconductors.	VASP
Matminer[204]	Python library for data mining and analyzing materials characteristics.	No specific package
MatCloud[362]	an HT computational platform for managing materials simulation data and resources.	VASP
MPInterfaces[363]	Python tool for HT DFT analysis of material interfaces.	VASP, LAMMPS
NOMAD[158]	infrastructure supporting the storage and analysis of computational materials science data.	FHI-Aims[364]
Pymatgen[365]	framework for automating physics simulations, from DFT to empirical pseudopotentials.	VASP, ABINIT, Gaussian
Qmpy[366]	backend tool for the OQMD, providing functionalities for managing computational data.	VASP
SEHC[367]	a framework for HT computing with automatic self-evaluation filtering.	VASP

accuracy. Starting with inexpensive, coarse filters—such as elemental abundance or rough DFT calculations—allows rapid elimination of clearly unsuitable candidates, thereby saving time and resources. This progressive funnel approach ensures that only the most promising materials are subjected to expensive, high-accuracy methods such as hybrid DFT or GW calculations. However, poor ordering can lead to unnecessary computational expenses if costly simulations are performed before practical constraints such as toxicity or synthesis feasibility. There is also a risk of false negatives if useful candidates are prematurely excluded during early filtering based on approximate descriptors. Conversely, false positives may persist into later stages if early filters are too lenient or if inaccuracies in high-level methods are not addressed. Overall, a carefully structured pipeline minimizes resource waste while maximizing the likelihood of discovering viable materials.[316]

HT workflow embodies several key principles: (i) a considerable time scale; (ii) a focus on data-driven discovery; (iii) the use of computational funnels; and (iv) the necessity for automation in HT methods. Each module within this workflow requires a degree of automation to enhance the efficiency of the HT methods.

Beginning with a large data set, HT calculations determine many materials properties through numerous DFT simulations.[340−342] This involves managing a big data project that includes acquiring, storing, analyzing, distributing, managing, and monitoring large data sets and HT DFT calculation tasks. To support this process, efficient and robust HT management tools are necessary.

Currently, 18 accessible HT programs exist that provide APIs to various DFT simulation packages,[316] including the Vienna Ab-initio Simulation Package (VASP),[343] Cambridge Sequential Total Energy Package (CASTEP),[344] Quantum Espresso (QE),[345] ABINIT,[346] and WIEN2k.[347] Table 1 lists the available HT screening packages.

ML for HT Screening and Computational Materials Design. Graph-based learners now furnish DFT-level descriptors at a fraction of the cost. Early models such as CGCNN, MEGNet, and ALIGNN first proved that formation energies, band gaps, and elastic constants could be inferred directly from atom-centered graphs without hand-engineered features.[197,198] Structure-agnostic transformers (CrabNet) later extended coverage to materials lacking a resolved crystal structure, as shown in Figure 9a.[368] State-of-the-art three-body networks, exemplified by M3GNet, compress months of plane-wave DFT into minutes while holding mean absolute errors below ≈40 meV/atom.[369] Domain-specific surrogates now proliferate: Materials Tensor (MatTen) predicts full elasticity tensors from geometry alone,[370] while deep convolutional neural networks, utilizing learned feature representations, enhance band gap prediction across the more than 20,000-entry QMOF database.[371]

Speed alone is insufficient; models must also decide where to compute next. DeepMind's GNoME coupled an uncertainty-weighted acquisition function to targeted DFT relaxations, expanding the catalog of convex-hull-stable crystals from $\approx 4.8 \times 10^4$ to 3.8×10^5 within a year.[199] The GNoME model is shown in Figure 9b. A similar classifier framework efficiently narrowed down the subset of van der Waals (vdW) dielectrics required for screening 2D field-effect transistors (FET) candidates, significantly reducing the computational expense.[372] In electrochemistry, an HT robotic platform guided by BO successfully identified a solvent system with a solubility of 6.20 M for 2,1,3-benzothiadiazole (BTZ) after evaluating only 218 out of over 2,000 candidates.[373] These virtual loops can already direct physical synthesis: Lawrence Berkeley National Laboratory's A-Lab realized 41 of 58 ML-ranked oxides in 17 days, demonstrating a closed computational−experimental cycle.[312] Follow-on studies apply equivalent protocols to perovskite solid solutions, sub-2 nm high-entropy catalysts and composite-cathode processing, each time compressing months of exploratory work to weeks.[374−376] Where screening is still constrained by preenumerated prototypes, diffusion-based generative models leapfrog enumeration. MatterGen more than doubled the novelty rate of earlier VAEs and GANs while maintaining a dynamic-stability filter.[245] The MatterGen model architecture is shown in Figure 9c. Equivariant DiffCSP produces symmetry-consistent crystals from random noise, and text-guided Chemeleon converts natural-language prompts into synthesizable structures, bridging materials design with foundation-language models.[377] For porous frameworks, ZeoDiff surpasses previous GAN models by more than 2,000 times in terms of structural accuracy when generating silica zeolites with specific target void fractions or Henry coefficients.[378] Generative pipelines are already application-ready: the Material Generation with Efficient Global Chemical Space Search (MAGECS) workflow combined a crystal diffusion VAE (CDVAE) with a population-based optimizer to propose alloy electrocatalysts, two of which achieved ≈ 90% Faradaic efficiency for $CO_2 \rightarrow CO$ after synthesis.[379] Diffusion models and generative approaches suggest

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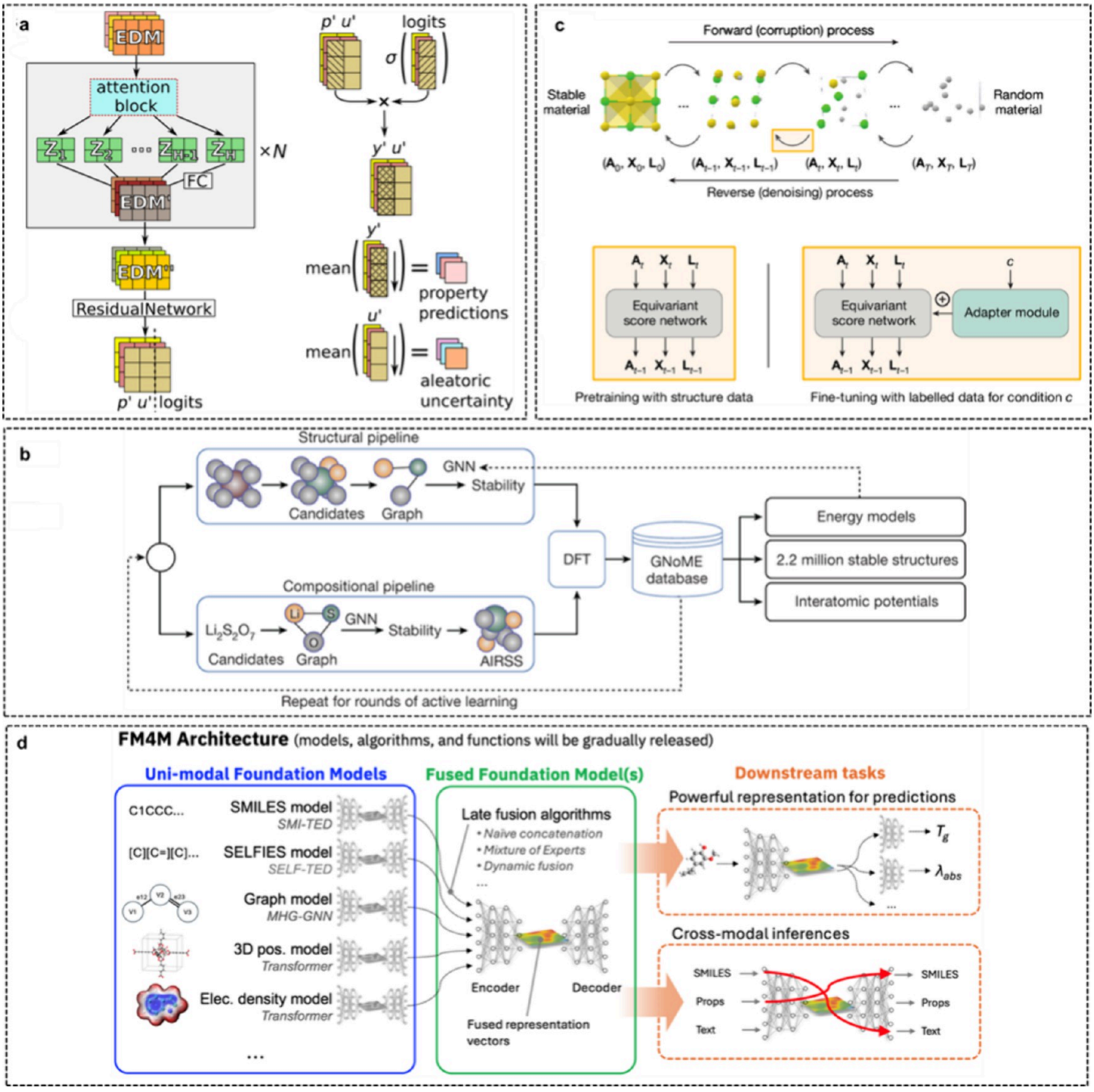


Figure 9. Foundation-model approaches that scale HT screening and computational materials design. (a) CrabNet: an element-derived matrix passes through stacked self-attention to give per-element contributions; their mean yields the predicted property and aleatoric uncertainty. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 368. Copyright 2021 The Author(s). (b) GNoME active learning loop: a GNN triages candidate structures/compositions, DFT validates the best, and the resulting 2.2 million-structure database now contains ≈3.8 × 105 new stable crystals and derived interatomic potentials. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 199. Copyright 2023 The Author(s). (c) MatterGen diffusion model: an equivariant score network reverses the corruption of atom types, coordinates, and lattices, fine-tuned via an adapter for property-conditioned generation. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 245. Copyright 2025 The Author(s). (d) FM4M suite: independently pretrained unimodal encoders (SMILES, graphs; 3-D, density) are late-fused into a shared latent that supports property prediction and cross-modal inference. Reprinted with permission under the terms of the Apache License 2.0. from ref 201. Copyright 2024 FM4M Contributors.

scalable routes for the design of multicomponent crystals, further accelerating materials discovery.[377,380]

Self-driving laboratories link ML inference, robotics synthesis, and real-time characterization into uninterrupted feedback loops. The Polymer Analysis and Discovery Array (PANDA) employs BO to optimize electrochromic polymer electrodeposition, completing campaigns that once took weeks in under 24 h.[381] Automated Reversible Addition−Fragmentation chain Transfer (RAFT) polymerization rigs and self-optimizing flow reactors similarly shorten reaction optimization from dozens of manual iterations to a handful of machine-selected experiments.[382−384] At the software layer, multimodal foundation models such as MultiMat and IBM's FM4M

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(shown in Figure 9d), unify text, graph, and 3-D inputs, enabling 'AI copilots' that can query literature, rank candidates, and dispatch robotic workflows from a single latent representation.[196,201] These advances point toward national-scale self-driving laboratory infrastructures in which computation, generative reasoning, and automated experimentation coevolve, radically accelerating the pace of materials innovation.

RECENT APPLICATIONS OF AI FOR MATERIALS DEVELOPMENT

Materials development is the iterative process of creating new materials or significantly modifying existing ones to design and produce materials that have the desired properties or functionalities for specific targeted applications. Grounded in synthesis, characterization, evaluation, and redesign, this process has been greatly accelerated by the integration of AI. As Kalinin et al. noted, AI enables high-resolution analysis of complex material systems by extracting patterns from imaging and spectroscopic data, which helps guide targeted improvements.[385] Raccuglia et al. further demonstrated that ML can learn from both successful and failed experiments, which will innovate materials development.[386] Building on this foundation, Abolhasani et al. argued that AI models can now drive experimental planning, perform real-time data interpretation, and guide robotic synthesis in closed-loop systems for the rapid development of synthesis or fabrication processes.[387] The following section explores recent applications of AI across three core areas of materials development: (1) materials characterization, where AI enhances image analysis, segmentation, and spectroscopic interpretation; (2) energy and battery materials, where models nondestructively predict composition and performance; and (3) autonomous experimentation, where robotic platforms and AI models work in tandem to accelerate synthesis, feedback, and optimization cycles. These advances highlight how AI is reshaping materials development into a faster, smarter, and more predictive enterprise.

Materials Characterization. The use of AI for microstructure and surface characterization has revolutionized the field, providing new capabilities for analyzing and understanding materials properties at the nanoscale.[388−390] AI techniques, particularly CNNs and other DL models, have automated traditional labor-intensive processes, improving the speed and accuracy of surface and microstructure analysis.[391]

A prominent example of AI application in this domain is fluorescence correlation spectroscopy (FCS).[392] By leveraging DL models like CNNs, researchers have reduced the amount of data required for molecular dynamics studies by up to 20 times. In the case of Tang et al., CNN-based models like FCSNet and ImFCSNet were used to accelerate the data analysis process, allowing for more rapid and accurate imaging of molecular interactions.[393] Also, Figure 10a shows that a CNN with U-Net architecture was employed to segment synchrotron tomography images of multiphase Al−Si alloys.[394] This approach significantly reduced the time required for segmentation and accurately identified complex microstructural features, such as silicon particles and interfaces between aluminides and silicon. The robustness and adaptability of the model were confirmed through its successful application to different alloy samples, showcasing its utility in diverse materials characterization scenarios.

Super-resolution techniques using DL models like Super-Resolution Residual Network (SRResNet) have been employed to enhance the resolution of low-resolution electron backscatter diffraction (EBSD) images.[395] Figure 10b shows the schematic of this technique, which allowed for more accurate analysis of local microstructural features, such as grain boundaries and stress distributions, facilitating better insights into materials behavior under mechanical stress. The method proved to be a fast, efficient, and reliable tool for enhancing microstructure images.

ML-based methods have also been applied to analyze microstructural image similarity in materials characterization. Techniques such as zero-normalized cross-correlation (ZNCC), mutual information (MI), and PCA effectively discerned similarities in microstructural images, aiding in understanding materials properties. As we can see from the results in Figure 10c, this approach can segment each phase in steel in both low and high-magnified microstructure images with a novel perspective on analyzing and comparing complex microstructures efficiently.[396]

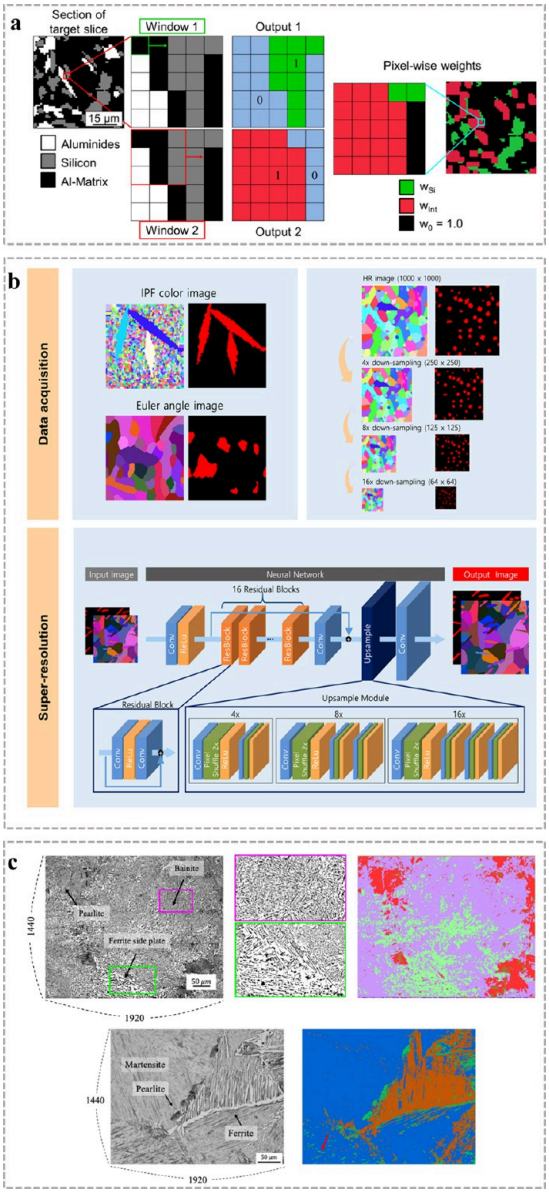


Figure 10. (a) U-Net architecture segments synchrotron tomography images, identifying silicon particles and aluminide interfaces in Al−Si alloys. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 394. Copyright 2019 The Author(s). (b) SRResNet improves the resolution of EBSD images for better analysis of grain boundaries and stress distributions. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 395. Copyright 2021 The Author(s). (c) ML techniques, such as ZNCC and PCA, segment steel microstructures at different magnifications, aiding in material comparison. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 396. Copyright 2020 The Author(s).

AI has also been integrated into atomic force microscopy (AFM) for surface profiling at the nanoscale.[397] Traditional AFM techniques face challenges due to tip convolution effects, which distort height profiles on small scales. To address this, ML models were applied to correct these distortions, improving the accuracy of AFM surface measurements. Bonagiri et al. highlights the use of DL models to

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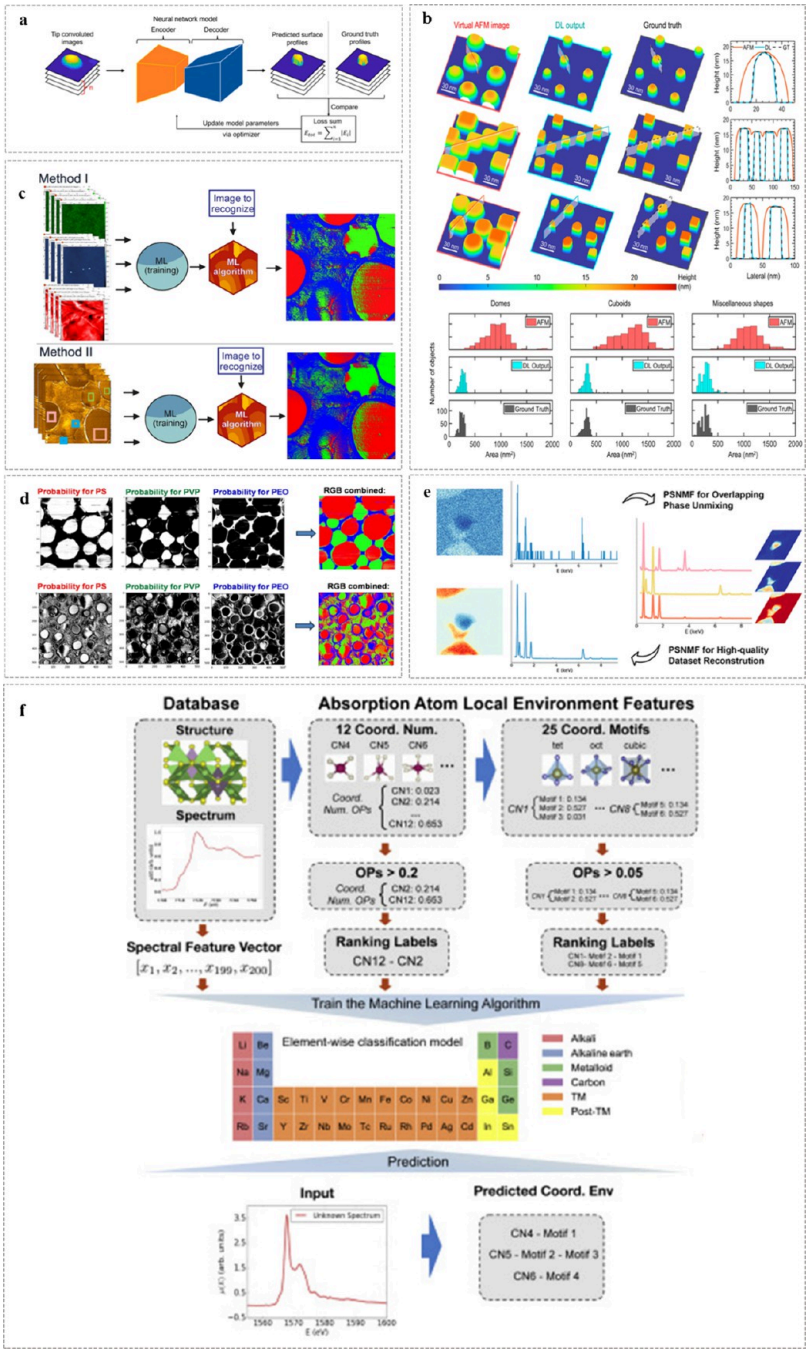


Figure 11. AI-enhanced techniques in nanoscale materials analysis and spectroscopy. (a) AI models correcting AFM surface profiling, improving height accuracy by addressing tip convolution. (b) AI-based AFM-MS for nanoscale materials classification in polymers and composites. Panels (a) and (b) are reprinted with permission from ref 398. Copyright 2024 American Chemical Society. (c) ML workflow for mapping polymer blends with enhanced resolution. (d) U-Net segmentation of porous materials linking microstructures to electrical conductivity. Panels (c) and (d) are reprinted with permission from ref 399. Copyright 2024 Elsevier Ltd. (e) AI-driven NMF and pan-sharpening for improved spectral and spatial resolution in STEM-EDX. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 406. Copyright 2024 The Author(s). (f) Random forest model for predicting coordination environments in XAS, enhancing the accuracy of atomic structure analysis. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 408. Copyright 2020 The Author(s).

enhance AFM accuracy by reducing tip convolution errors.[398] Figure 11a,b depicts the differences in surface profiling before and after the application of AI-based corrections, showing a significant improvement in profile accuracy and reduced surface area deviations.

In addition to AFM, AI has been utilized in mechanical spectroscopy with AFM (AFM-MS)[399] to achieve high-resolution materials identification at the nanoscale, with a lateral resolution of 1.6 nm. Using neural networks and random forests, the method classifies polymeric and composite materials based on AFM data.[399] Figure 11c illustrates the workflow, while Figure 11d highlights the technique's ability to map polymer blends, surpassing the precision of traditional spectroscopic methods. This AI-driven approach enhances predictive modeling by linking nanostructures to macroscopic properties, which is crucial for the design of advanced nanocomposites and polymer systems.

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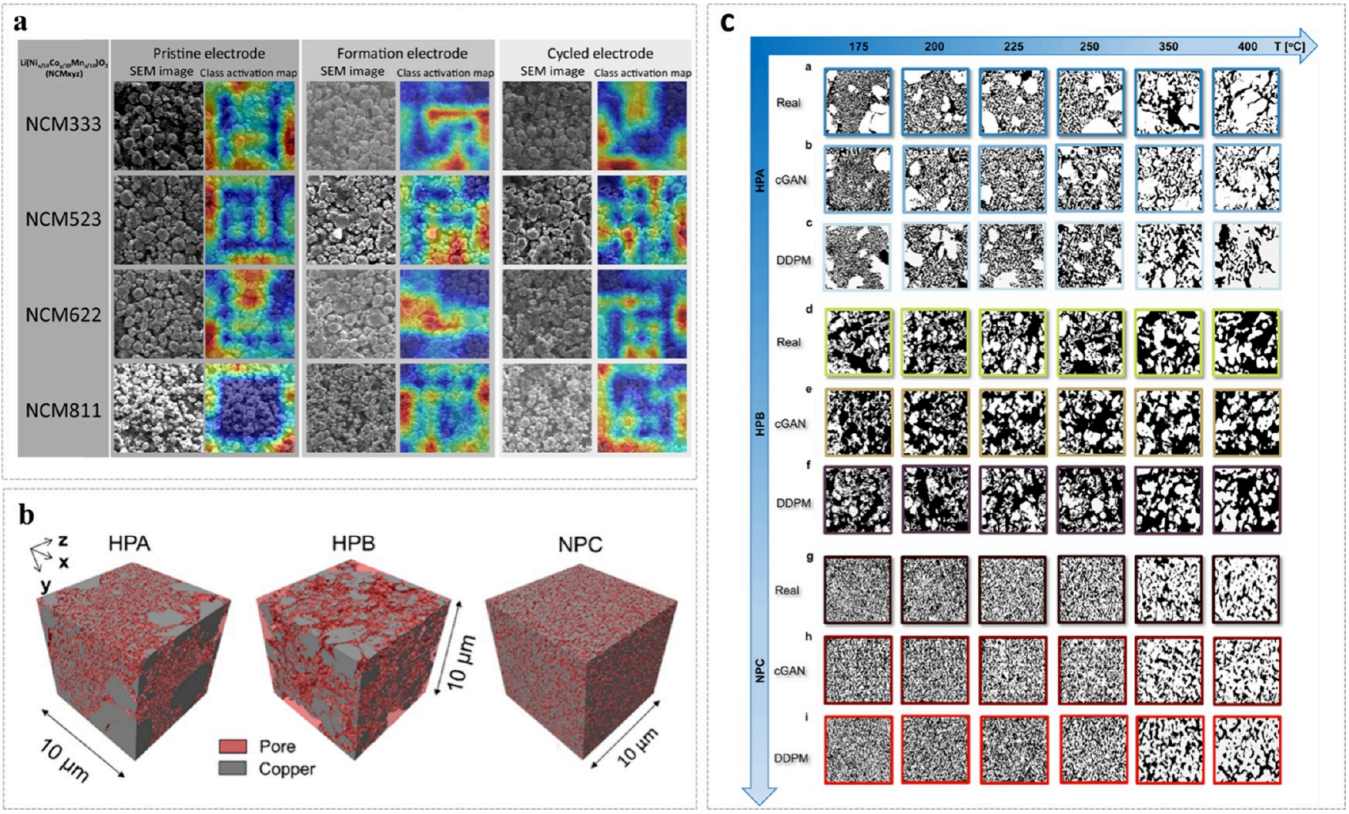


Figure 12. AI applications in battery materials and energy storage characterization. (a) CNN predicts the composition and cycling state of the LiBs. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 128. Copyright 2024 The Author(s). (b) U-Net segments porous copper microstructures for conductivity analysis. (c) Predicted conductivity of porous copper for optimizing material performance. Panels (b) and (c) are reprinted with permission under a Creative Commons CC BY 4.0 License from ref 409. Copyright 2024 The Author(s).

Building on AI-driven approaches for materials characterization, multicrystalline informatics combines experimental, theoretical, computational, and data science techniques to decode complex materials phenomena. Applied to multicrystalline silicon, this approach uses ML models, such as artificial neural network (ANN), to predict properties like crystal orientation distribution and carrier recombination velocity based on microstructure, defect, and grain boundary data.[400,401] By providing insights into the relationship between grain boundary structures and materials behavior, this methodology improves the accuracy of property prediction and facilitates the optimization of growth processes.[402]

The automation of these traditional manual processes through AI-driven tools has enabled more detailed and accurate materials analysis. From enhancing nanoscale surface measurements to automating the segmentation of complex microstructures, AI techniques are driving advancements in materials characterization that were previously unachievable.[403]

AI has significantly contributed to X-ray and spectroscopy analysis by automating complex data interpretation tasks and improving the precision of materials property predictions. ML models, such as CNNs and random forests, are increasingly used to process and analyze large data sets generated by X-ray and spectroscopy techniques.[404] These AI-driven methods have enhanced data analysis speed and increased materials insights' accuracy and depth.

A key example of AI in X-ray analysis is the application of non-negative matrix factorization (NMF)[405] combined with pan-sharpening techniques in scanning transmission electron microscopy with energy-dispersive X-ray spectroscopy (STEM-EDX). This work employed these techniques to improve nanoscale materials analysis's spatial and spectral resolution, specifically in studying nanocatalysts. The combination of NMF and pan-sharpening significantly enhanced phase separation and material contrast, allowing researchers to identify and characterize nanoscale features with greater clarity. Figure 11e illustrates the improvements in spectral accuracy and material contrast achieved through AI-driven sharpening algorithms, showcasing the benefits of applying AI to STEM-EDX data.[406]

Ghose et al. employed neural network ensembles (NNE) to predict molecular X-ray absorption spectra (XAS), incorporating uncertainty quantification (UQ) to enhance prediction reliability.[407] This approach improves spectral accuracy and provides pointwise uncertainty estimates, making it particularly effective in handling out-of-sample data and structural perturbations. The model can generalize larger and more complex molecular systems, and its uncertainty-aware framework enables robust materials characterization under varying structural and chemical conditions. This AI-driven method offers a valuable tool for real-time materials characterization, significantly improving efficiency in X-ray spectroscopy analysis by combining accurate predictions with confidence metrics.[407]

In XAS, AI has been applied to streamline the analysis of complex data sets. Figure 11f illustrates the method of using the random forest to predict the coordination environments of materials based on XAS data, achieving an accuracy of over 85%.[408] This work highlights how this model was trained on a large data set of XAS measurements to accurately identify atomic coordination environments, which are crucial for understanding materials' chemical bonding and structure. This demonstrates the strong correlation between the predicted and actual coordination environments and the model's ability to perform highly accurate XAS analysis with minimal human intervention.[408]

By incorporating AI into X-ray and spectroscopy analysis, researchers can extract detailed insights from vast and complex data sets more quickly and accurately than ever before. ML models in these fields not only automate traditional manual processes but also open

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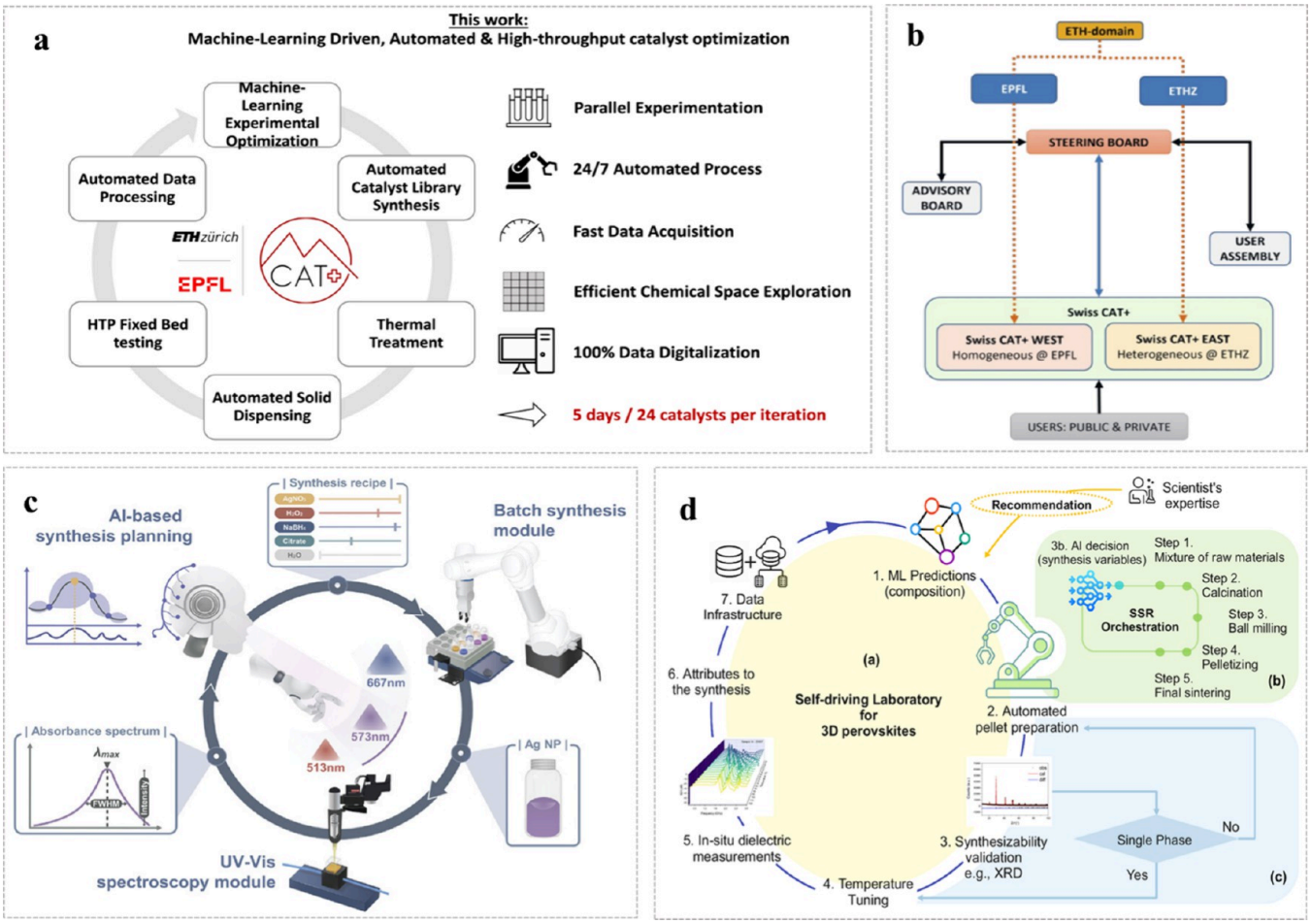


Figure 13. AI-driven catalyst discovery and synthesis optimization. (a) ML optimization for the CO_2 hydrogenation catalysts. Adapted with permission from ref 420. Copyright 2023 Elsevier Inc. (b) Swiss CAT+ platform for catalyst discovery using ML. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 421. Copyright 2023 The Author(s). (c) AI-guided silver nanoparticle synthesis via BO. Adapted with permission under a Creative Commons CC BY 4.0 License from ref 422. Copyright 2024 The Author(s) (d) AI-driven discovery of perovskite materials for energy applications. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 375. Copyright 2024 The Author(s).

new possibilities for discovering material properties that might otherwise go unnoticed.

AI has become an essential tool for accelerating materials development in energy materials and battery research. Using ML models, researchers can automate the analysis of complex materials data, improving the efficiency of materials property predictions and enabling HT experimentation. AI techniques, particularly CNNs and DL models, have been applied to characterize energy materials' composition, structure, and performance, including those used in battery technologies.[128]

One of the most impactful applications of AI in this field is using CNNs to predict the composition and cycling states of LiB cathodes. Traditionally, this type of analysis requires destructive testing, which is time-consuming and labor-intensive. In Oh et al., CNN models were applied to SEM images of LiB cathodes, allowing for the nondestructive prediction of their composition and state of charge.[128] The CNN model successfully classified the different phases present in the cathodes, significantly reducing the need for manual image interpretation. Figure 12a showcases the results of the CNN-based classification, illustrating how the model outperformed traditional manual methods by providing more accurate and detailed predictions of the cathode composition.[128]

Another significant advancement in AI for energy materials characterization is seen in U-Net models applied to porous copper, commonly used in electronic devices and energy applications. In Wijaya et al., U-Net DL architectures were used to segment the complex microstructures of porous copper and predict its electrical conductivity.[409] The AI model achieved an impressive segmentation accuracy of 94%, allowing researchers to link microstructural features to key electrical properties. Figures 12b,c display the segmented microstructures and the corresponding predicted conductivity values, providing crucial insights into the materials performance under various conditions.

By integration of AI into the characterization of energy materials and batteries, researchers can perform faster and more accurate evaluations of materials properties, which are critical for improving the performance and longevity of next-generation energy storage systems. The ability to predict materials properties without extensive physical testing represents a significant leap forward in the development of efficient, high-performance energy materials.

AI for Autonomous Experimentation. Recent advances in AI have transformed materials discovery from a trial-and-error process into a data-driven and autonomous enterprise.[410,411] Enabled by predictive modeling, optimization algorithms, and robotic execution, self-driving laboratories span the entire workflow—from hypothesis generation to synthesis, characterization, and feedback-based refinement.[312] These paradigms speed up the process and improve the precision of the outcomes.[412] To provide a clearer understanding of these developments, we organize this into two major themes: (1) the core methodologies that enable autonomous experimentation and (2) application cases—particularly in catalysis—where self-driving laboratories have demonstrated practical success.

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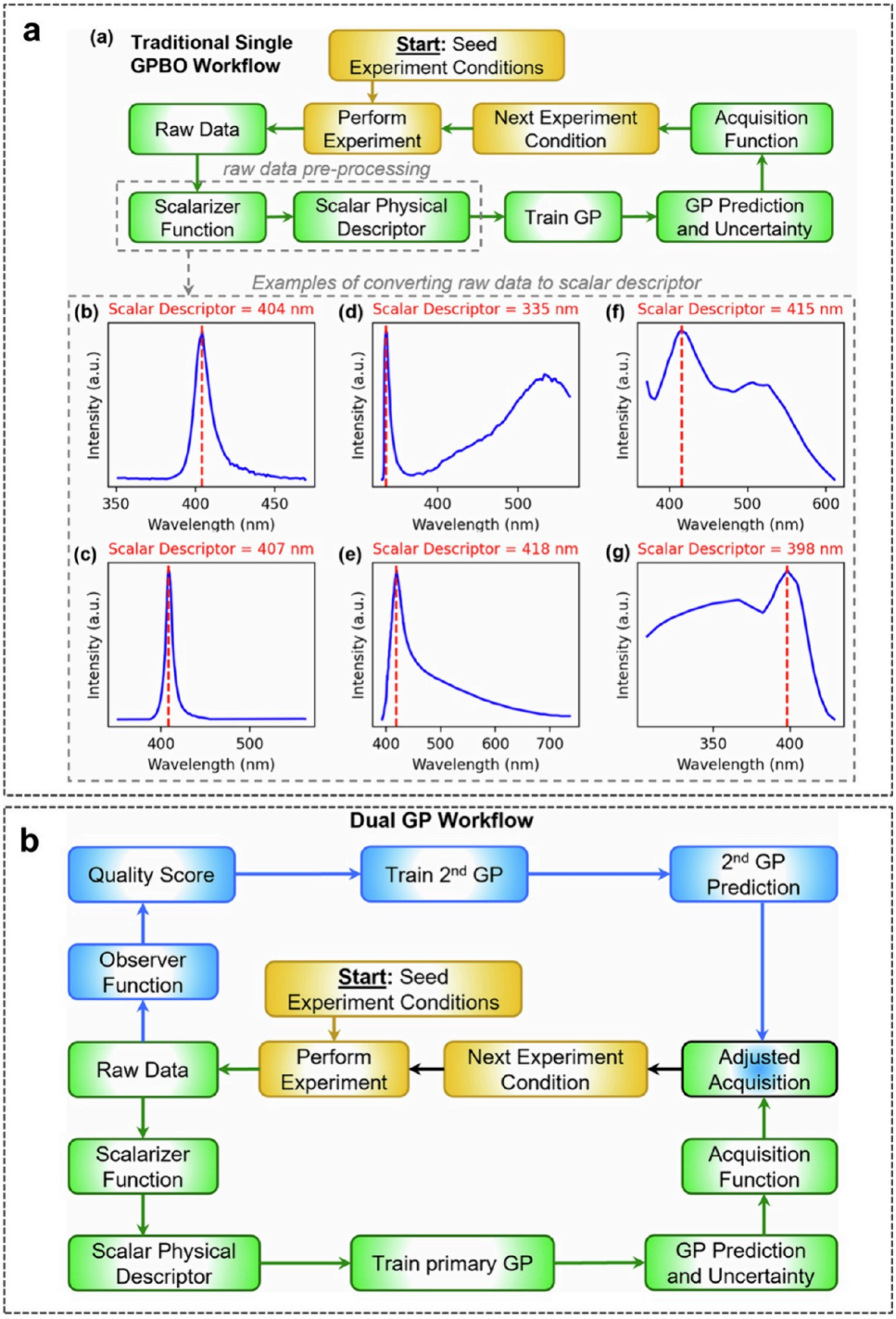


Figure 14. Comparison of the traditional single GPBO workflow and the dual GP workflow, illustrating the impact of scalar descriptor quality on optimization. (a) The conventional GPBO process, where variations in the scalarizer function can lead to misleading descriptors. (b) Examples of PL spectra showing how different scalarizers capture or miss crucial spectral features, highlighting the need for a second GP to assess descriptor quality and refine the exploration space. Panels (a) and (b) are reprinted with permission under a Creative Commons CC BY 4.0 License from ref 423. Copyright 2025 UT-Battelle, LLC.

LLMs have proven to be effective in extracting synthesis knowledge from unstructured literature and supporting inverse design. For instance, ChatGPT for Chemistry achieved 90−99% precision in parsing over 26,000 Metal−Organic Framework (MOF) synthesis parameters, enabling ML models to predict crystallization outcomes with 87% accuracy.[413] Similarly, ChatMOF leveraged GPT-4 to generate novel MOFs with desired properties through an integrated design pipeline.[414] Transformer-based NLP systems like A-Lab recommended synthesis conditions from over 30,000 papers, successfully synthesizing 35 out of 58 candidates.[312] LLMatDesign further extended LLMs into closed-loop optimization, reducing optimization steps by over 50% and achieving formation energies as low as −2.72 eV/atom through self-reflection mechanisms, demonstrating efficient *in silico* optimization.[415] Autonomous robotics plays a key role in executing AI-guided experiments, particularly when integrated with real-time characterization.[416] For example, a multi-robot system developed by Cooper et al. performed 12 distinct tasks—including PXRD and crystal handling—achieving a 24/7

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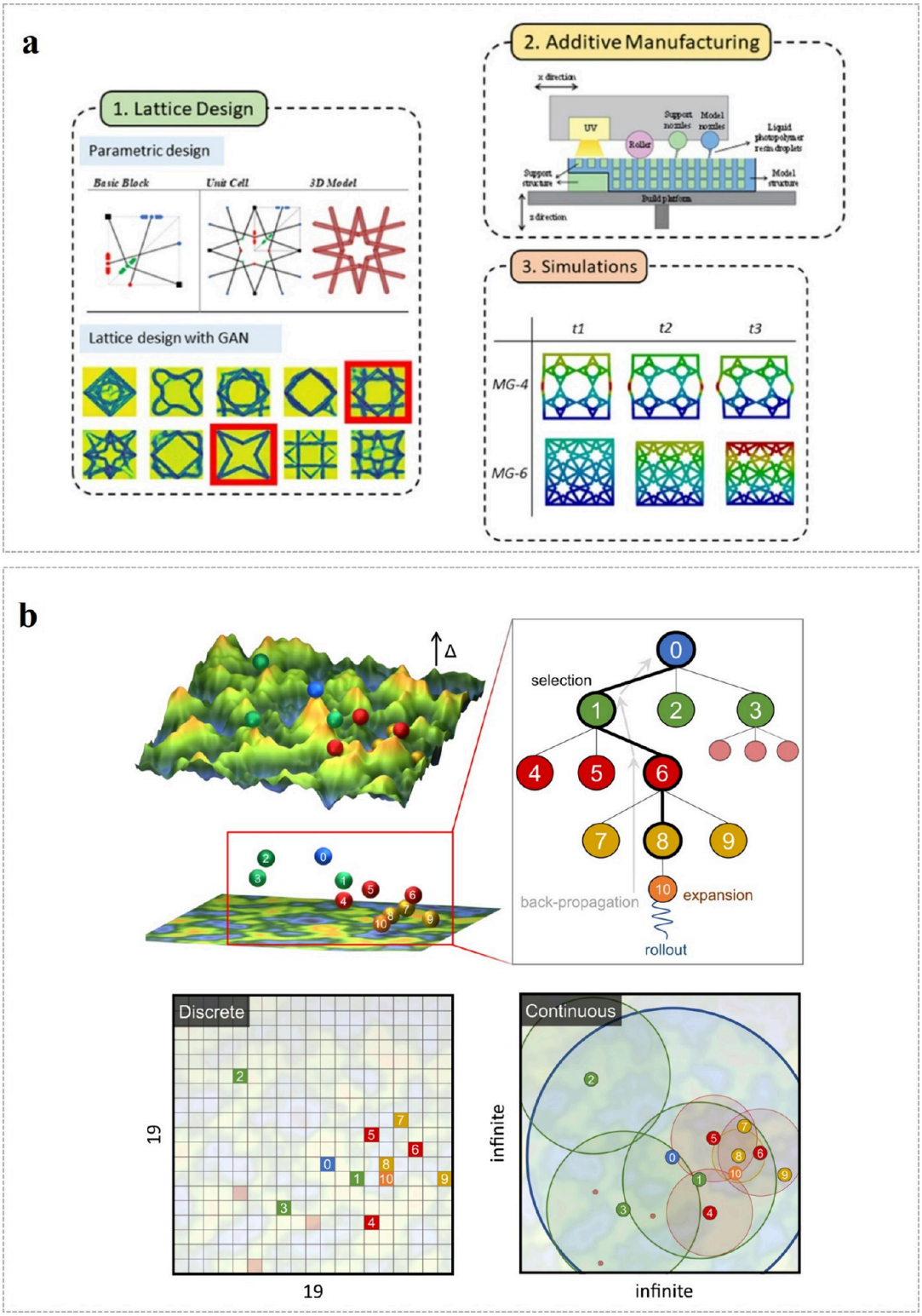


Figure 15. Workflow of GAN for designing high-strength lattice structures. (a) Illustration of the generative process for creating lattice structures optimized for additive manufacturing, demonstrating significant improvements in strength and elongation. Adapted with permission from ref 439. Copyright 2024 Elsevier Ltd. (b) Integration of MCTS with RL for optimizing synthesis parameters, highlighting the exploration of high-dimensional materials design spaces and computational efficiency. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 446. Copyright 2024 The Author(s).

throughput of 168 samples per week, four times higher than manual workflows. To interpret experimental outputs, intelligent agents like the Crystallography Companion Agent (XCA) apply CNN ensembles to diffraction data, achieving over 95% accuracy across chemically diverse systems such as $BaTiO_3$ and Ni−Co−Al alloys.[417] XCA also integrates multimodal inputs such as EDX and PXRD, enabling robust decision-making under uncertainty. These developments exemplify the synergy between robotics and ML-based analysis, closing the loop

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from prediction to interpretation and laying the groundwork for fully autonomous discovery pipelines.

In parallel, hybrid strategies combining BO with reinforcement or active learning have further improved the experimental efficiency. For example, the study by Kalinin et al. demonstrated that integrating BO with reinforcement learning through BO-guided exploration, while RL prioritized promising candidates based on prior outcomes, enabled accelerating structure−property optimization in microscopy-driven experiments by up to 5 times compared to random or grid-based search to discovery new materials.[412] Similarly, CAMEO combined physics-informed active learning with real-time phase mapping to discover an optimal Ge−Sb−Te phase-change composition in just 19 experiments, achieving an E_g of 0.76 eV—three times greater than standard compositions. This framework exemplifies how prior physical knowledge and real-time feedback can be jointly leveraged to refine hypotheses and reduce experimental burden.[418] To further enhance the explore/exploit balance in acquisition, recent work has proposed threshold-driven hybrid acquisition policies, demonstrating superior performance over traditional single acquisition functions in materials discovery tasks.[419]

AI has fundamentally transformed catalyst development by enabling autonomous experiments and HT synthesis. Researchers can now rapidly explore vast chemical spaces and optimize catalytic performance with minimal human intervention. In CO_2 hydrogenation, Ramirez et al. utilized a BO-driven platform to identify high-performance catalysts with minimal experimental trials (Figure 13a).[420] The Swiss CAT+ project combined HT screening with ML to optimize both homogeneous and heterogeneous catalysts (Figure 13b).[421] A BO-based closed-loop system also enabled the autonomous synthesis of silver nanoparticles with improved size control and quality (Figure 13c).[422] Furthermore, robot-assisted platforms efficiently explored optimal perovskite solid solution compositions for energy applications by balancing stability, efficiency, and environmental resilience (Figure 13d).[375] Collectively, these examples highlight how AI-driven experimentation is reshaping catalyst discovery and synthesis optimization exploration, while RL allows it to prioritize promising candidates based on prior outcomes. This approach significantly reduced the time needed to discover new materials compared with traditional random or grid-based search methods.

The traditional Gaussian process BO (GPBO) workflow starts with seed experimental conditions followed by experiments and preprocessing, where raw data are converted into scalar physical descriptors using a predefined scalarizer function.[121] These descriptors, along with experiment conditions, form a training set to train a GP surrogate model, which then uses an acquisition function to determine the next experimental conditions based on predictions and uncertainties. However, variations in scalarizer quality can significantly impact optimization, as some descriptors capture key features while others miss crucial information, leading to misleading results.[423] To address this, a second GP is introduced as an observer, assessing the compatibility of the scalarizer function with raw data and assigning a quality score to constrain the exploration space.[423] This dynamic constraint filters out low-quality scalarizers and irrelevant data, refining the BO workflow by guiding experiments toward informative regions while avoiding noisy or infeasible spaces. Figure 14 compares the traditional single GPBO workflow and the dual GP workflow, highlighting these limitations and the proposed improvement.[423] In Figure 14a, the conventional GPBO process follows a sequence of experiment execution, data processing, GP training, and acquisition function-driven exploration, but inconsistencies in the scalarizer function can lead to unreliable descriptors. Figure 14b shows examples of scalar descriptors from raw experimental data, illustrating how photoluminescence (PL) spectra from hybrid organic−inorganic compounds can yield varying descriptor quality. While some descriptors (*e.g.,* 404 and 407 nm) capture essential spectral features, others (*e.g.,* 335 and 415 nm) fail to represent crucial information due to scalarizer inconsistencies, emphasizing the need for quality assessment to ensure meaningful scalar descriptors in the GPBO workflow.

In addition to materials discovery, AI has been employed for autonomous reaction mapping. A prime example is the Fast-Cat system described in Bennett et al., which autonomously mapped Pareto fronts for gas−liquid reactions using BO.[424] The system continuously updated its models in real-time based on experimental results, allowing it to adjust reaction parameters such as temperature, pressure, and reactant concentration to optimize reaction rates and product yields. By focusing on Pareto optimization, Fast-Cat was able to find an optimal balance between competing objectives in complex chemical reactions.

These AI-driven systems enable faster exploration of chemical spaces and allow for more efficient reaction optimization by reducing experimental waste and providing deeper insights into the underlying reaction mechanisms. By autonomously mapping complex reactions and optimizing materials properties, AI is paving the way for a new era of data-driven experimentation, where human intervention is minimal and the speed of discovery is dramatically increased.

RECENT APPLICATIONS OF AI FOR MATERIALS OPTIMIZATION

Materials optimization is the strategic refinement and fine-tuning of an existing material or material system to achieve the best possible performance within the given constraints. Unlike materials discovery, which searches for new materials or materials development, which creates new materials, optimization focuses on enhancing the properties or processing parameters of a material that has already been developed or identified. AI has significantly accelerated both materials design optimization (*e.g.,* structure generation and candidate selection) and materials process optimization (*e.g.,* synthesis planning and parameter tuning) by enabling the rapid exploration of design spaces, real-time control, and faster convergence. For example, RL frameworks like AlphaFlow automate complex synthesis,[425] and BO with active learning reduce experimental effort while maintaining accuracy in systems like batteries and redox flow cells.[426,427] The following section explores these AI-driven strategies across both design and process optimization, highlighting their role in accelerating and automating the materials optimization workflow.

Materials Design Optimization. AI has become a transformative tool in materials design optimization. AI has accelerated materials design optimization by employing advanced ML techniques, data analytics, and computational modeling.[428−437] These innovations allow for optimizing materials properties with unprecedented speed and precision. AI-driven approaches are increasingly integrated across various stages of materials research, from high-throughput screening and predictive modeling to experimental automation and real-time process control, opening new possibilities for creating advanced materials that align with modern technological demands and sustainability goals.[373]

One of the most impactful applications of AI in materials design involves GANs. In materials optimization, GANs can be used to create new materials structures or compositions that exhibit desired properties.[254,438] This approach leverages GANs' ability to learn complex data distributions and generate new samples similar to the training data. For instance, GANs have been successfully employed to design high-strength lattice structures for additive manufacturing, resulting in materials with enhanced mechanical properties.[439] Figure 15a illustrates the workflow of generating these lattice structures using GANs, highlighting the process from design to testing, where significant improvements in strength and elongation were achieved compared to traditional lattice structures. In addition, coupling GANs with RL, specifically the actor-critic (AC) method, leads to efficient exploration of large design spaces. This integration enables the discovery of novel materials with tailored properties, significantly reducing the need for extensive experimental trials. Such a hybrid approach accelerates the discovery process while ensuring optimal materials performance.[440,441]

On the other hand, multi-objective optimization has also been used to optimize materials design.[442] For example, Synthesis Planning and Rewards-based Route Optimization Workflow (SPARROW) has been

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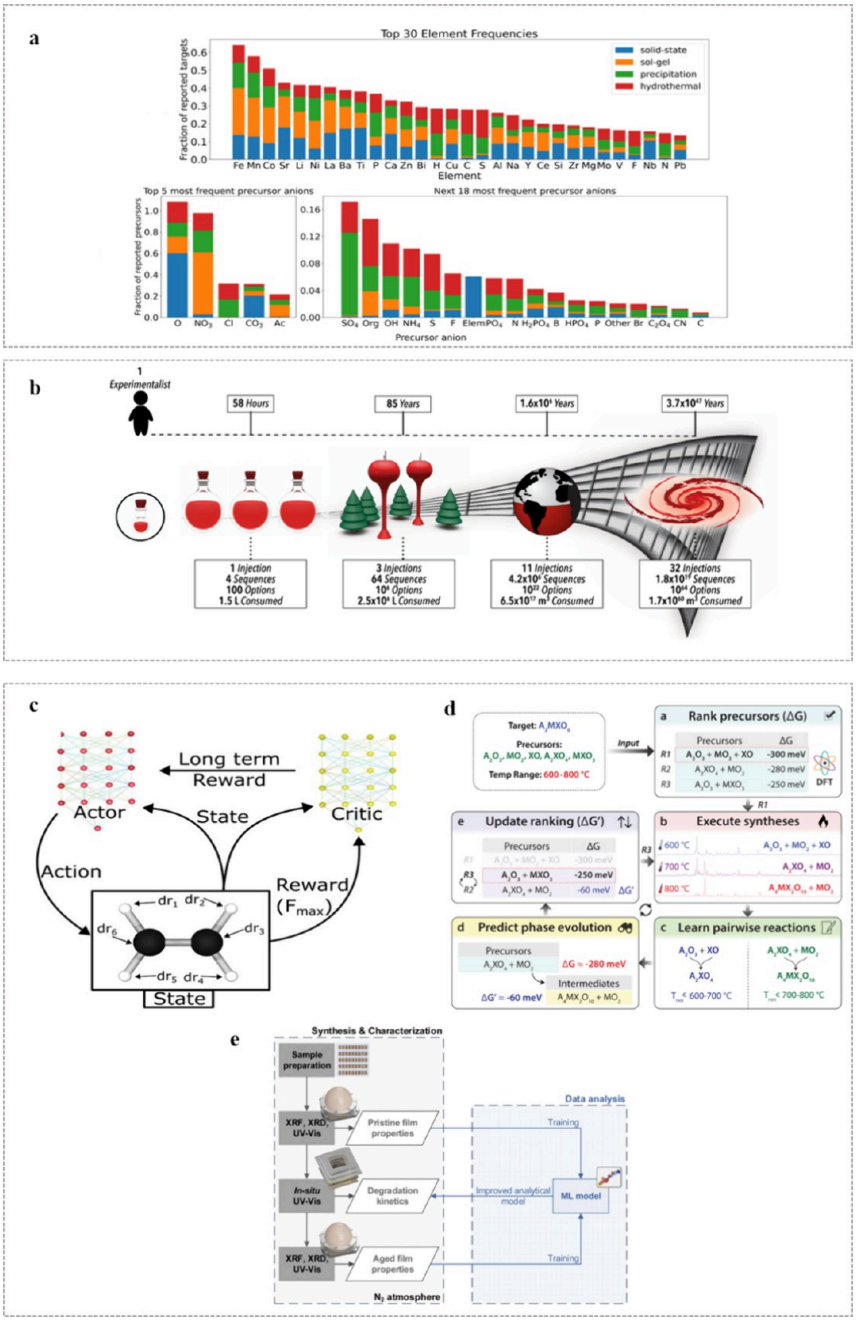


Figure 16. AI-driven optimization for materials process parameters. (a) Supervised ML model predicting synthesis conditions and revealing the relationship between the precursor composition and process conditions. Reprinted with permission from ref 461. Copyright 2023 American Chemical Society. (b) RL feedback loop in AlphaFlow autonomous system for optimizing multistep chemical reactions. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 425. Copyright 2023 The Author(s). (c) Actor−critic process in the RL where the actor adjusts atomic positions, the critic evaluates rewards, and the policy is optimized iteratively. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 463. Copyright 2024 The Author(s). (d) ARROWS^3 optimization of precursor selection, significantly reducing trials for discovering optimal synthesis routes. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 468. Copyright 2023 The Author(s). (e) ML is used to analyze changes in the optical spectra. Reprinted with permission under a Creative Commons CC BY 3.0 License from ref 469. Copyright 2024 The Royal Society of Chemistry.

employed to design molecules.[443] The SPARROW selects an optimal subset of molecules from a candidate pool (from virtual libraries, *de novo* design, or expert source) by weighing each candidate's utility (*e.g.,* property predictions) against the cost of synthesizing them, considering reaction likelihoods and synthetic feasibility. It reduces synthesis costs by using overlapping intermediates and reaction steps, ultimately improving efficiency while maintaining high model confidence in the synthetic success. Moreover, another SPARROW version, which is called the improved SPARROW search algorithm (ISSA), has been utilized to predict the state-of-health (SOH) of LiBs accurately and efficiently using robust indicators from partial discharging data.[444]

RL plays a key role in optimizing material synthesis by iteratively adjusting synthesis parameters to achieve optimal conditions. An example of this is the minimum thermodynamic competition (MTC) concept, which is integrated with RL to minimize kinetic byproducts in materials synthesis. This approach ensures that the desired materials phase remains thermodynamically stable compared to

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competing phases, leading to materials with improved purity and properties.[445] In materials optimization, RL is further enhanced by integrating it with the Monte Carlo tree search (MCTS), a heuristic search algorithm used to improve decision-making in high-dimensional environments. MCTS combined with RL enables efficient exploration of complex multidimensional problems, such as optimizing synthesis parameters or designing materials properties.[446] Figure 15b showcases how MCTS is used to explore high-dimensional materials design spaces, significantly improving prediction accuracy and reducing computational costs.

AI's impact on materials design is further enhanced through the integration of ML with robotics. This combination has led to the development of automated materials synthesis systems that optimize multiple properties simultaneously, drastically reducing the trial-and-error approach traditionally used in materials development. For example, a recent study integrated AI with robotics to accelerate the development of biodegradable plastic substitutes.[447,448] By utilizing a robotic system, researchers could conduct HT experiments and optimize the composition of nanocomposite materials, improving both environmental sustainability and materials performance. Integrating AI, ML, and robotics represents a significant leap toward fully automated materials design, where experimental conditions are dynamically adjusted based on real-time data, further improving the efficiency of materials discovery.[373,449]

AI techniques such as active learning have proven invaluable in materials design by reducing the experimental burden. Active learning iteratively improves model predictions by guiding experimental choices, focusing on the most informative data points, and maximizing the efficiency of the design process.[450−452] For instance, a study on iron−chromium redox flow batteries demonstrated that active learning, combined with ML algorithms like gradient boosting, achieved prediction accuracies with a R^2 greater than 0.92, significantly surpassing traditional methods. Specifically, active learning reduced the required experimental data points by over 60% compared to conventional approaches while maintaining prediction accuracies within ± 0.15% for energy efficiency and ± 0.8% for capacity.[427]

Materials Process Optimization. AI is rapidly transforming materials process optimization by making the design, synthesis, and manufacturing of materials more efficient and effective.[453−459] AI techniques such as ML, RL, generative models, and BO are employed to discover new materials, optimize processing conditions, predict materials properties, and guide experimental workflows. This reduces the reliance on trial-and-error experiments, accelerates materials development, and lowers costs in materials science. In this section, we discuss the specific applications of these AI techniques in materials process optimization.

One promising AI technique is supervised ML, which has been applied to optimize inorganic reaction processes.[449] Oliynyk et al. applied supervised ML to accurately predict crystal structures, significantly expediting materials synthesis.[460] Supervised ML models can classify synthesis routes (*e.g.,* solid-state, sol−gel, hydrothermal, and precipitation methods) based on compositional features and predict critical reaction conditions such as calcination and sintering temperatures. For instance, Karpovich et al. developed an ML-driven framework that uses a conditional VAE (CVAE) to predict distributions of synthesis conditions (*e.g.,* temperature and time.)[461] This probabilistic approach surpassed traditional point-based methods, offering enhanced flexibility and precision across various synthesis scenarios. Additionally, the framework employed SHapley Additive exPlanations (SHAP) to identify key features impacting synthesis parameters, enhancing interpretability and practical utility.[386] Zhou et al. reviewed ML methodologies specifically targeting inorganic materials synthesis, highlighting predictive accuracies for synthesis conditions.[462]

The framework's effectiveness in process parameter optimization is demonstrated in Figure 16a, which captures key trends, such as the relationship between precursor composition and synthesis conditions. This allows for enhanced control over morphology, yield, and purity, streamlining experimental workflows and providing a more data-driven approach to inorganic materials synthesis.[461]

RL plays a significant role in autonomous materials process optimization by making dynamic, sequential decisions based on real-time feedback.[463] Zhou et al. explored RL applications in chemical reactions, demonstrating RL's potential to efficiently optimize experimental workflows autonomously.[464] Shields et al. developed AlphaFlow, an RL-based autonomous system designed for optimizing complex, multistep chemical reactions.[465] Specifically, Stein et al. reported AlphaFlow's successful optimization in colloidal atomic layer deposition (cALD) processes, achieving up to 450% greater efficiency than traditional methods by autonomously adjusting reagent sequences and phase separations.[466] One notable application of RL is in autonomous systems like AlphaFlow, which is designed for the optimization and discovery of complex, multistep chemical reactions.[425,467] Without human intervention, the system can optimize multiple reaction parameters, including reagent sequences and phase separations.

Real-time decision-making and automation are key to AlphaFlow's success, as depicted in Figure 16b, which shows the RL-guided feedback loop integrated into modular fluidic components. As mentioned above, the system outperformed traditional cALD methods, as evidenced by improvements in the reaction efficiency and photoluminescence intensity of core−shell Quantum Dots (QDs). These advancements demonstrate how RL can significantly accelerate and enhance multistep chemical synthesis.

RL also has applications in optimizing molecular transition states and minimum energy pathways (MEPs).[463] A recent study used an actor−critic RL framework to explore molecular structure space and efficiently locate transition states in chemical reactions.[463] This model significantly reduced the number of computational steps required to optimize reaction pathways compared to traditional methods. Figure 16c shows the actor−critic process in the RL where the actor generates an action that modifies the atomic positions within the molecule. The molecule's current arrangement represents the state. A reward is determined based on the new configuration and is provided to the critic, which estimates the long-term reward. This estimate, along with the current state, is used to refine the actor's policy. With the updated policy, the cycle repeats.[463] Figure 16d visualizes the RL workflow for molecular optimization. In this case, reactions using different precursor sets (R1, R2, and R3) are ranked based on their driving force, determined through DFT calculations. Experiments at progressively higher temperatures help identify reaction intermediates with phases confirmed by XRD. Pairwise reaction temperatures (T_{rxn}) and resulting products are extracted from experimental data. These identified reactions allow the prediction of intermediates for other precursor sets, leading to updates in their remaining driving forces. The precursor rankings are then refined accordingly. The chemical formulas shown are placeholders, with no restrictions on compositions for ARROWS^3 applications.[468] In addition, the RL is used to analyze changes in optical spectra, helping to link these changes to phase transformations during aging, depicted in Figure 16e.[469] The inert-gas synthesis and characterization workflow for combinatorial perovskite libraries includes a combinatorial perovskite library, an inert-gas XRD Dome, and a climate chamber for UV−vis aging experiments. Characterization data from pristine and aged thin films are collected and used to train an ML model, which analyzes optical spectra from in situ UV−vis aging experiments.[469]

Another advancement in materials process optimization is the use of AI-driven robotic systems to enhance solid-state materials synthesis.[470] The ARROWS^3 algorithm is a key development in this area, optimizing precursor selection to increase the thermodynamic driving force of reactions while avoiding byproducts.[470] Applying ARROWS^3 to over 200 synthesis experiments, the algorithm successfully reduced the number of trials needed to discover optimal synthesis routes, outperforming other optimization methods like BO and genetic algorithms.[470]

ARROWS^3's thermodynamic precursor ranking system provides a more efficient workflow, and this demonstrates its superior performance in discovering optimal synthesis routes with fewer experi-

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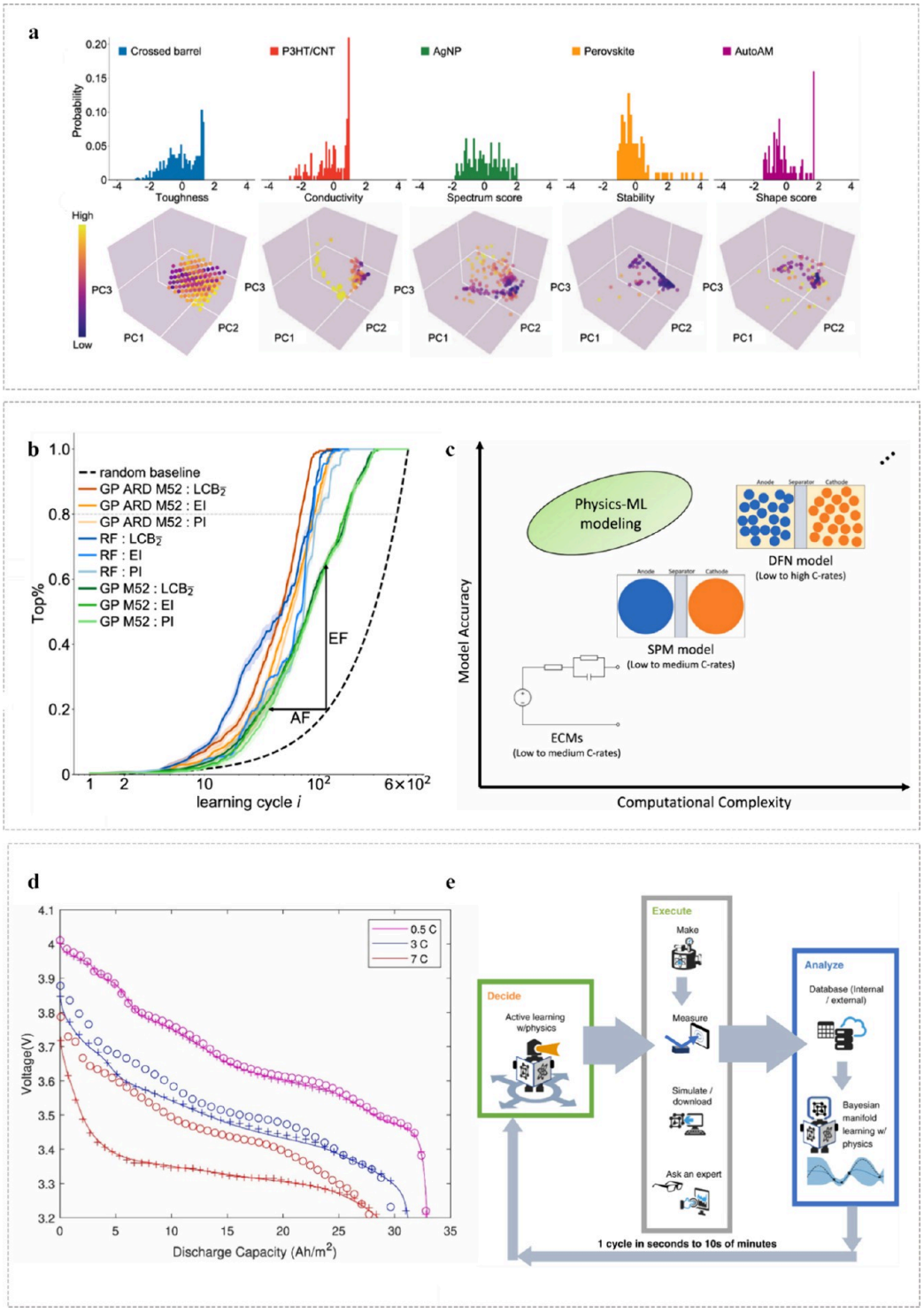


Figure 17. BO in materials design and energy storage technologies. (a) Visualization of experimental data sets used for benchmarking BO algorithms across multiple domains. (b) Comparison of battery degradation over time using AI-driven hybrid models. Panels (a) and (b) are adapted with permission under a Creative Commons CC BY 4.0 License from ref 426. Copyright 2021 The Author(s). (c) Comparison between hybrid and traditional models for voltage prediction in lithium-ion batteries. (d) Performance of BO models in navigating high-dimensional spaces for optimizing material properties. Panels (c) and (d) are adapted with permission from ref 467. Copyright 2023 Elsevier Ltd. (e) CAMEO workflow. Reprinted with permission under a Creative Commons CC BY 4.0 License from ref 418. Copyright 2020 The Author(s).

ments.[468] These findings highlight how AI-driven systems can streamline complex chemical processes, accelerating the discovery of high-purity materials with reduced experimental workloads.

BO is another crucial AI technique for materials process optimization. BO efficiently guides experimental workflows by leveraging surrogate models and acquisition functions to navigate complex design spaces with fewer iterations. A comprehensive study by Zhang et al. benchmarks BO across various materials science domains, demonstrating its ability to optimize material properties in an HT experimental setting.[467,471]

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The power of BO in materials process optimization can be found in the study that compared surrogate models, such as GPs with anisotropic kernels and RFs, where GPs with anisotropic kernels provided superior accuracy in navigating high-dimensional design spaces.[426] Figure 17a visualizes the diverse experimental data sets used in this study, and Figure 17b highlights the performance of BO algorithms across these data sets, with GPs consistently outperforming other models.[426] This study highlights the benefits of BO in streamlining experimental workflows, especially by minimizing the need for expensive and time-intensive experiments.

AI is also making strides in the optimization of materials processes for energy storage technologies. One significant application is in the design and management of LiBs. Hybrid frameworks that integrate physics-based models, such as the single particle model with thermal (SPMT) dynamics and equivalent circuit models (ECMs), with ML algorithms have been developed to optimize battery performance.[467,471]

These hybrid models, which combine physical models with feedforward neural networks (FNNs), achieve higher accuracy in voltage predictions across a wide range of charging and discharging currents (C-rates).[467,471] The study by Wu et al. shows that these hybrid models reduce prediction errors by up to 90% compared with traditional models. Figure 17c illustrates the comparison between hybrid and traditional models, and Figure 17d demonstrates the hybrid model's ability to maintain accuracy even at high C-rates.

Finally, AI is advancing the field of closed-loop autonomous systems for materials exploration. The Closed-loop Autonomous Materials Exploration and Optimization (CAMEO) system integrates Bayesian active learning to optimize phase mapping and materials property evaluation, significantly reducing the number of required experiments.[418] Specifically, the CAMEO methodology has demonstrated up to a 10-fold reduction in required experiments compared to exhaustive exploration, with each iteration taking mere seconds to minutes depending on the experimental task. CAMEO can dynamically adjust experimental conditions based on real-time data by selecting the most informative experiments. In this case, the autonomous cycle starts by importing data from multiple databases, such as material composition information from the composition spread and computational materials data from the AFLOW DFT database,[155] which is then analyzed using physics-informed Bayesian ML to predict structural and functional properties for materials without existing data, along with associated uncertainties. Physics-informed active learning then determines the most informative material to investigate in order to meet user-defined goals. It selects the next sample for characterization through autonomous control of the high-throughput X-ray diffraction system at a synchrotron beamline or, if needed, requests input from a human operator. The workflow of CAMEO is shown in Figure 17e. This approach exemplifies how AI-driven systems can accelerate the optimization of advanced materials, balancing exploration and exploitation in complex materials synthesis processes.

CHALLENGES AND LIMITATIONS

Data Quality and Representation. Materials science uses diverse data sets, including experimental results, computational outputs, and literature-derived data.[472,473] These data sets often suffer from inconsistencies, incomplete annotations, and variable quality, which undermine the reliability of the AI model.[474,475] Additionally, integrating heterogeneous data from multiple sources remains a significant hurdle.[476]

Interpretability of AI Models. Advanced AI models, such as DL architectures, excel in predicting materials properties and behaviors but often lack interpretability.[391] This 'black-box' nature limits their application in materials research, where understanding mechanisms and validating predictions are essential.[477] This lack of transparency hinders user trust and limits adoption in high-stakes applications.

Multi-modal and Multi-output Integration. The ability to predict multiple materials properties or integrate diverse data types—such as microscopy images, spectral data, and textual descriptions—is still emerging.[95,478] Training these complex models requires large, curated multimodal data sets often unavailable in materials science.

Causality Challenges. Current AI models primarily identify correlations between materials structures, properties, and performance.[479,480] However, understanding causality is critical in materials design, as it enables mechanistic insights and predictive generalizations.[481]

Computational Cost. Training and deploying sophisticated models, such as GenAI or LLM models for materials science, require extensive computational resources,[482] which can limit access to smaller research groups.

Ethical Concerns and Data Security. Proprietary or sensitive materials data from industries and research organizations pose challenges in data sharing and collaboration.[483] Risks like data poisoning or model tampering raise concerns about reliability and trustworthiness in materials science applications.

FUTURE TRENDS AND DIRECTIONS

We all witness GenAI's transformative power and capabilities in today's world. GenAI is undoubtedly revolutionizing every aspect of life including research, creativity, productivity, and problem-solving. Particularly in the materials science domain, the advanced GenAI models, such as VAEs,[484] GANs,[133] diffusion models,[135] and transformers,[137] along with LLMs, including OpenAI's GPT,[485] DeepSeek,[486] LLaMA,[487] Anthropic's Claude,[488] Gemini,[489] and the like, are poised to play a pivotal role in shaping the future of materials research. These technologies can potentially revolutionize fundamental materials data analysis, autonomous experimentation, hypothesis generation, and the prediction or generation of materials' structures and properties. Their unparalleled ability to integrate diverse data sets, analyze complex material systems, and derive knowledge representation from materials language positions them as essential tools for driving innovation and accelerating progress in materials science. Below are some aspects where generative AI will help advance future AI/ML–driven materials research.

Causal Mechanistic Models. The integration of causality into LLMs helps to overcome their reliance on statistical correlations and spurious patterns, enabling models to focus on true causal relationships. Techniques such as debiased token embedding and counterfactual data augmentation allow LLMs to identify causal dependencies, reduce biases, and improve interpretability and reliability.[490] In materials science, such causal reasoning can be applied to develop causal mechanistic models that analyze the relationships between synthesis conditions and materials properties. For example, such models can potentially identify how changes in the synthesis temperature or doping concentration impact materials performance, enabling accurate hypothesis validation before physical experiments.

Multi-modal and Multi-output Predictions. Generative models enable multimodal and multi-output predictions.[491,492] These models excel at managing various prediction tasks within a single platform, offering capabilities that enable a flexible, comprehensive, and in-depth understanding of the prediction results.[493,494] In materials science, this model type will hold immense potential by integrating different modalities

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of materials data, such as spectral data from spectroscopy, image data from microscopy, and textual information from the scientific literature, into a unified framework for tackling complex material property and structure predictions. Additionally, their capability to generate multiple targets simultaneously enables concurrent predictions of material properties and structures, providing deeper and more comprehensive insights into the prediction outcomes.

Direct Use of Raw Materials Data. Recent studies highlight the transformative potential of LLMs in performing regression tasks directly from unstructured textual data through the innovative approach of in-context learning (ICL).[495,496] With the versatility of strings and tokens to represent diverse data types, it becomes essential to explore how LLM embeddings can extend to nontabular inputs, such as graphs, images, and videos, unlocking their potential across complex and varied data modalities.[494,497] This capability is particularly beneficial in the materials domain, where it reduces information loss and minimizes the need for extensive feature engineering and selection compared to structured tabular data. By leveraging LLMs to process unstructured data and embeddings for nontabular formats, researchers can streamline materials discovery, optimize synthesis processes, and predict complex properties with minimal preprocessing.

Autonomous Materials Research Platforms. LLMs are pivotal for autonomous research platforms, acting as the intelligent core for experimental design, analysis, and decision-making.[498] By integrating literature analysis, real-time data interpretation, and predictive modeling, LLMs enable closed-loop systems that optimize experiments and reduce trial-and-error.[499] They can potentially facilitate human-AI-robotics collaboration through natural language interfaces and streamline workflows from automating literature reviews and experimental planning to optimizing materials synthesis and characterization, ultimately accelerating the path to materials product development. Equipped with the capability to detect errors, adapt to outcomes, and suggest innovative approaches, LLMs enhance the efficiency and creativity of materials research, ultimately accelerating discoveries and advancing materials product development.

Physics-Informed AI Model. Hybrid models combining physics-based frameworks with generative AI bring the best of both worlds—scientific rigor and data-driven innovation. The future of AI-driven materials research lies in integrating physics-based knowledge with ML, as exemplified by physics-informed neural networks (PINNs). These models incorporate physical laws, such as partial differential equations, into their framework, enabling accurate predictions even with limited data.[500] Physics-informed VAEs,[501] GANs,[502] diffusion models,[503] transformers,[504] and LLMs[505] can integrate physical constraints into generative processes, ensuring realistic simulations of materials behavior. For example, they can predict new stable material phases while adhering to known thermodynamic laws, bridging gaps between theory and experimentation.

Explainable AI (XAI) for Materials Science. XAI is essential for advancing materials research by addressing the critical need for transparency and scientific understanding in AI-driven predictions.[506] There have been several XAI platforms created by the world's leading companies, such as Google's What-If Tool (WIT),[507] Microsoft's InterpretML,[508] IBM's AIX360,[509] Amazon SageMaker,[510] Facebook's Captum,[511] and NVIDIA Clara.[512] In materials science, where researchers frequently handle complex, high-dimensional data sets, identifying key features and mechanisms is crucial for validating results and informing experimental design. Tailored XAI frameworks equipped with interpretable models and visualization tools are crucial in bridging the gap between AI predictions and real material behaviors. These tools not only enhance transparency and trust but also facilitate interdisciplinary collaboration by making complex AI outputs understandable and actionable.

Integration with Quantum Computing. The integration of quantum computing with AI is poised to revolutionize materials science. Quantum computers' ability to perform complex calculations at an unprecedented speed opens up new avenues for simulating material behaviors at the quantum level. Combining quantum algorithms with AI models, such as deep learning networks, could allow for faster and more accurate predictions of material properties, particularly those that involve quantum mechanical effects. This synergy will enable the development of materials with entirely new properties that were previously unimaginable. Quantum ML (QML) models, leveraging quantum mechanics and AI, are expected to significantly enhance materials discovery by simulating quantum systems and solving problems that classical computers cannot tackle efficiently.[513−516]

Human-AI Collaboration and Ethics. As AI technologies advance, it is crucial to explore the ethical implications and establish frameworks for responsible human-AI collaboration. The potential for AI to support or even replace human decision-making in materials science raises important questions regarding bias, transparency, accountability, and decision-making authority. Ethical considerations must guide the development of AI technologies to ensure they are used in ways that benefit society as a whole rather than reinforcing existing inequities. Ensuring transparency in AI models and making them interpretable will be key in fostering trust between humans and machines, especially when AI is used to support critical decision-making in materials design and innovation.[517−520]

CONCLUSIONS

AI, ML, and DL have revolutionized materials science by enabling rapid materials discovery (structure generation and identification, property prediction, high-throughput screening, and computational design), materials development (characterization and autonomous experimentation), and materials optimization (design and process optimization). These technologies have accelerated research workflows, reduced the reliance on trial-and-error approaches, and facilitated the design of materials with tailored properties for next-generation applications. Autonomous laboratories, multimodal frameworks, and emerging GenAI models have further streamlined research by integrating diverse data types and automating complex workflows.

However, significant challenges remain. The scarcity and heterogeneity of high-quality data sets, the nontransparent nature of many AI models, and the difficulty in integrating multimodal data hinder broader adoption. The lack of standardized data ecosystems, interpretable AI frameworks, and causal inference models limits trust and applicability, particularly in high-stakes materials research. Addressing these issues requires efforts to develop physics-informed AI, hybrid models combining scientific rigor with data-driven innovation, and XAI frameworks. Ethical concerns, data security, and

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computational demands of sophisticated models also need attention to ensure equitable access and reliability.

Looking ahead, future efforts will focus on improving robustness, integrating causal reasoning and physics-informed AI, and leveraging multimodal models to enhance scalability and transparency, unlocking new opportunities for more advanced materials discovery, development, and optimization. The integration of quantum computing with AI will enable faster and more accurate material property predictions, and ethical frameworks will ensure responsible human-AI collaboration, addressing concerns of bias, transparency, and accountability in decision-making. Emerging GenAI models such as VAEs, GANs, diffusion models, and transformers, as well as LLMs, hold immense potential to transform materials science further. These models can integrate multimodal data, generate multi-output predictions, and directly utilize raw materials data, streamlining discovery and optimization. Autonomous research platforms powered by LLMs are promising to enable closed-loop experimentation, real-time analysis, and collaborative human-AI workflows. Furthermore, advancements in causal mechanistic modeling and physics-informed frameworks will deepen our understanding of materials behaviors, bridging the gap between theory and experimentation.

By addressing current challenges and leveraging these advancements, AI-driven materials science can tackle global challenges and meet the demands of future technologies, enabling breakthroughs in energy storage, catalysis, sustainability, and beyond. The integration of AI with human expertise will drive a new era of innovation and discovery in the field.

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Notes

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VOCABULARY

Bayesian Optimization (BO), A probabilistic optimization method used to find the best solution of expensive or complex black-box functions with minimal evaluations. BO strategically decides which point in the search space to sample next, making it highly efficient for tuning material synthesis or processing parameters.; **Deep Learning (DL)**, An advanced branch of ML that relies on multilayered neural networks to automatically extract high-level features from large, complex data sets. It excels in handling unstructured data such as images, text, and spectra, making it particularly effective in materials characterization.; **Generative Models**, AI models designed to create new data samples resembling those from a training data set. Techniques such as variational autoencoders, generative adversarial networks, and diffusion models can generate novel material structures or microstructures that exhibit desired properties.; **Graph Neural Networks (GNNs)**, A family of neural network architectures that process and learn from data structured as graphs—particularly useful for modeling atomic or molecular systems. By capturing the relationships between nodes (*e.g.*, atoms) and edges (*e.g.*, bonds), GNNs enable accurate property predictions and insight into complex materials.; **Machine Learning (ML)**, A subset of AI in which algorithms learn patterns directly from data rather than following explicitly programmed rules. ML encompasses various techniques—including supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning—to predict or classify outcomes and reveal hidden insights.; **Materials Informatics**, A specialized field at the intersection of materials science, data science, and AI that focuses on gathering, managing, and analyzing materials data. It harnesses advanced computational methods to streamline the discovery and optimization of new materials.; **Transfer Learning (TL)**, A technique where knowledge gained from one domain or task is reused to improve learning performance in another, related domain—especially useful when data are limited. TL helps reduce the need for extensive, domain-specific data sets while maintaining robust predictive accuracy.

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